Are Frontier VLM Agents Ready to Be Robot Generalists?

An Empirical Study with the Embodied Agent Arena

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Abstract. Frontier vision-language models (VLMs) increasingly estimate scenes, ground interactions, and generate executable actions. How far these native capabilities support embodied generalism across diverse tasks remains unclear. We introduce Embodied Agent Arena to assess eight VLM agents across Geometry, Spatial Reasoning, Affordance, Task Planning, and Manipulation. The arena contains 1,000 cases drawn from 32 established sources and GeoProbe, our new benchmark for geometric estimation on Blender renders and real-scene images. A minimal harness preserves source observations and operations while leaving perception, reasoning, and action selection to the model. Separate measures of metric precision, functional grounding, and native goal completion connect local competence to complete task outcomes. Astra leads geometric estimation and planning, while Opus achieves the highest measured affordance overlap, contact validity, and Baseline manipulation success. These local strengths coexist with endpoint errors in tracing and low household-task completion. Supplementary comparisons of richer observations and multi-round review show model- and task-dependent effects. Current VLM agents thus fall short of embodied generalism: they often make partial progress without satisfying all task goals within allotted time and interaction budgets.

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1 Introduction

A robot generalist must turn scene understanding into actions that satisfy a complete goal. Locating an object, selecting a usable contact, and executing a grasp do not establish that the object reaches its destination. Vision-language models (VLMs) increasingly address spatial relations (Ma et al., 2024, Yang et al., 2024) and action-relevant grounding (Wang et al., 2025a, Song et al., 2024). Learned visuomotor policies connect observations to robot actions (Kim et al., 2024, Black et al., 2024), while language-model controllers select skills or generate executable programs (Ahn et al., 2022, Liang et al., 2022). Frontier agents increasingly combine these mechanisms with perception support and visual action rehearsal (Fu et al., 2026, Guo et al., 2026, Li et al., 2026b, Zhang et al., 2026a). This progress raises a central issue: which robotic requirements can the VLM's own perception, reasoning, and action selection satisfy?

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Figure 1. Overview of Embodied Agent Arena. Each row illustrates six tasks in one capability domain, spanning camera and scene estimation, spatial relations, functional grounding, goal-directed planning, and robot manipulation. Fixed-input probes test estimates and judgments; interactive episodes test goal completion. Spatial-tracing overlays show reference paths and a 3D target region.

Assessing generalism requires connecting individual capabilities to complete task performance. Aggregate scores can hide errors in metric scale, reference frames, or placement conditions; terminal failure can conceal useful progress. This distinction motivates our central question: *Are frontier VLM agents ready to be robot generalists?*

We introduce Embodied Agent Arena to evaluate eight VLM agents across Geometry, Spatial Reasoning, Affordance, Task Planning, and Manipulation (Figure 1). Its 1,000 cases combine 32 established sources with GeoProbe, which varies camera motion, object motion, and scale references to diagnose geometric estimation. A minimal harness retains each source's observations and operations while leaving task decisions to the model. Separate measures of geometric precision, functional grounding, and native task completion reveal both cross-capability strengths and the requirements left unsatisfied within a task.

The results expose a gap between local precision and complete execution: Astra produces the closest paths yet often misses their endpoints, and its local control strengths coexist with low household-goal completion. Supplementary experiments examine richer observations and multi-round review, showing that additional evidence has model- and task-dependent effects. Our contributions are:

- **An evaluation perspective centered on task requirements.** We distinguish local estimates and interaction evidence from complete goal satisfaction across five domains.

- **A unified arena with a new geometric benchmark, GeoProbe.** We combine 32 established sources with 168 GeoProbe cases, using controlled Blender scenes and real images to diagnose camera, motion, depth, and scale estimation.
- **A diagnosis of the gap from local competence to complete tasks.** We compare eight agents and identify the task requirements that remain unsatisfied despite strong geometric precision and functional-contact localization.

2 Related Work

Embodied evaluation has developed along two complementary directions. Control suites test execution and task generalization (James et al., 2019, Mees et al., 2021), whereas visual diagnostics isolate spatial inference (Ma et al., 2024, Yang et al., 2024) and interaction grounding (Song et al., 2024, Qian & Fouhey, 2023). Scene and task question answering (Majumdar et al., 2024, Sermanet et al., 2023) extends these diagnostics to embodied decisions (Li et al., 2024b, Qi et al., 2025). The former establishes whether a goal is achieved; the latter resolves particular perceptual or reasoning requirements. Evaluating both in the same agents connects local competence to complete task outcomes.

The interface determines how that competence reaches execution. Learned policies map observations and instructions to actions (Brohan et al., 2023, Zitkovich et al., 2023), increasingly across tasks (Ghosh et al., 2024, Kim et al., 2024, Black et al., 2024); language-mediated control instead selects supplied skills (Ahn et al., 2022, Huang et al., 2023b) or composes executable programs (Liang et al., 2022, Singh et al., 2023). Programs organize control primitives and dependencies (Fu et al., 2026, Du et al., 2025), geometric feedback (Huang et al., 2023a), and reusable policy interfaces (Li et al., 2026a). Their decisions still depend on visual evidence quality (Li et al., 2025, Wu et al., 2026) and temporal consistency (Huang et al., 2026b, Zhao et al., 2026b,a). Demonstrations (Jiang et al., 2023, Huang et al., 2026a), retained histories (Zhang et al., 2026b), and execution verification (Guo et al., 2024, Huang et al., 2026c) address these dependencies. Task success therefore depends on model decisions and interface support.

Broader evaluations connect perception with goal-directed interaction (Yang et al., 2025a, Cheng et al., 2025). Table 1 compares their coverage, diagnostics, and scale: OpenEQA (Majumdar et al., 2024), Embodied Arena, and Hard Vision, Easy Vision combine visual probes with navigation (Ni et al., 2025, Rasheed et al., 2026); VLABench also evaluates manipulation (Zhang et al., 2024). Embodied Agent Arena connects geometry, functional grounding, and native task completion across 1,000 cases, with GeoProbe isolating camera, motion, and scale requirements. This unified evaluation measures the current level of VLM agents' native embodied generalist capabilities across tasks, from local judgments to task completion.

3 The Embodied Agent Arena

3.1 Benchmark Construction and Capability Domains

Data construction. Embodied Agent Arena contains 1,000 cases from 32 established sources and one new benchmark, GeoProbe, organized into five capability domains (Figures 1–2). Each case specifies observations, a query or goal, operations, and scoring. We introduce GeoProbe to diagnose the metric ambiguities behind geometric estimates. Its 168 cases vary camera motion, object motion, and scale references in controlled Blender scenes, complemented by

Table 1. Evaluation coverage, diagnostic analysis, and sample scale. Fixed probes use supplied observations; region probes score image grounding; manipulation requires arm/gripper control; memory spans sessions. Calib./pose tests camera calibration or relative pose; Forms denotes task-type/condition breakdowns. Scale counts QA items or task episodes; NR: aggregate count not reported (Appendix G).

	Sample scale	Evaluation coverage					Diagnostic analysis	
Benchmark	Instances	Fixed	Region	Nav.	Manip.	Memory	Calib./pose	Forms
OpenEQA (2024)	1,636	✓	✗	✓	✗	✗	✗	✓
VLABench (2024)	NR	✓	✗	✗	✓	✗	✗	✓
BEAR (2025)	4,469	✓	✓	✗	✓*	✗	✗	✓
EmbodiedBench (2025a)	1,128	✗	✗	✓	✓	✗	✗	✓
EmbodiedEval (2025)	328	✗	✗	✓	✗	✗	✗	✓
Embodied Arena (2025)	NR	✓	✓	✓	✗	✗	✗	✓
CaP-X (2026)	700/tier	✗	✗	✓	✓	✗	✗	✓
WorldLines (2026b)	331	✓	✗	✗	✗	✓	✗	✓
Hard/Easy Vision (2026)	NR	✓	✓	✓	✗	✗	✗	✓
Embodied Agent Arena	**1,000**	✓	✓	✓	✓	✗	✓	✓

✓ included; ✗ not evaluated; *BEAR-Agent online validation. Scale details: Appendix G.

real images. For interactive cases, the recorded terminal state connects the agent's action sequence to the task's goal conditions. Appendix A gives source allocation and adaptations.

Geometry. Given images, the agent estimates intrinsics, pose, depth, dimensions, or displacement. InFlux and Map-free test calibration and relative pose (Liang et al., 2025, Arnold et al., 2022); GeoProbe varies camera motion and scale references. TraceSpatial-Bench adds paths on 56 scenes (Zhou et al., 2026), testing reference-trajectory error alongside start, arrival, and collision constraints. These targets separate numerical precision from geometric feasibility.

Spatial Reasoning. The agent infers relations, viewpoints, configurations, and temporal order from images or video (Ma et al., 2024, Wang et al., 2025b, Yang et al., 2024). Examples include comparing object distances, imagining another observer's view, and inferring camera motion. Each question specifies a reference frame and expects a category, quantity, or point set.

Affordance. Given a scene and intended action, the agent predicts a contact point or bounding box. UMD tests tool functions such as cutting and scooping (Myers et al., 2015); instruction- and action-conditioned 3DOI queries test interaction contacts (Qian & Fouhey, 2023). Point validity tests whether the contact lies in the functional region; overlap measures region coverage.

Task Planning. The agent organizes source-provided operations into a sequence that achieves a household, navigation, or scientific goal (Shridhar et al., 2019, Wang et al., 2022), using visual, textual, or symbolic feedback. Compound goals require several terminal conditions, such as heating an object and placing it at the destination. For terminal state s_T and required predicates $g_1, \ldots, g_K$, complete success is

$$\mathrm{Success}(s_T, g) = \mathbf{1}\left[\bigwedge_{k=1}^{K} g_k(s_T)\right]. \tag{1}$$

Manipulation. The agent executes robot motions or skills for placement, insertion, articulation, and coordinated handling (James et al., 2019, Chen et al., 2025a). Language-conditioned variants change the target relation or object (Zhang et al., 2024); sequential tasks require preserving

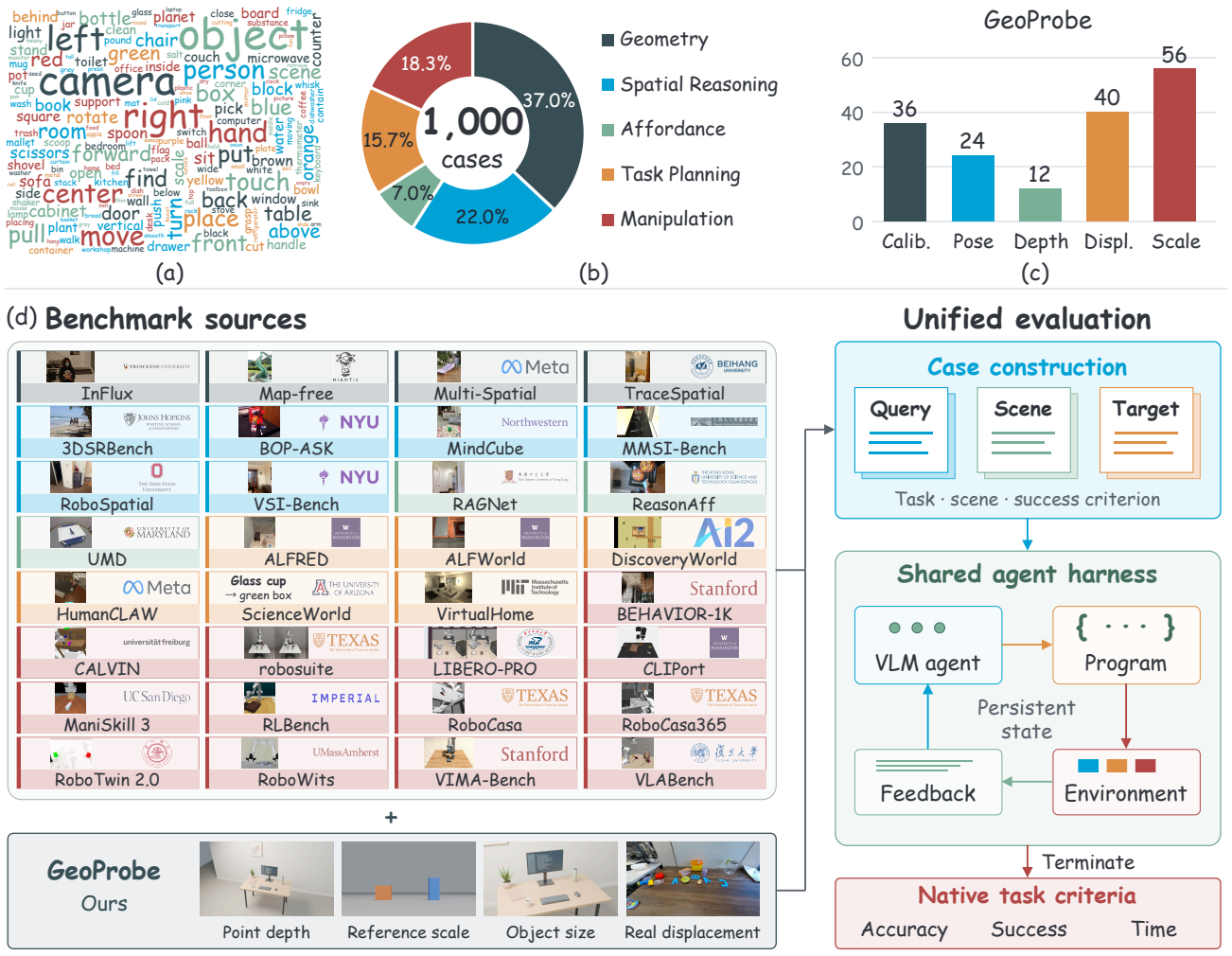

Figure 2. Construction of the 1,000-case arena. (a) Representative task vocabulary across sources. (b) Five-domain composition, including 112 spatial-tracing cases in Geometry. (c) GeoProbe estimation targets. (d) Source queries, scenes, and targets connect model interaction to termination and scoring. Colors identify domains; logos denote source institutions.

progress across ordered goals. Native success checks task completion, while condition-level scores record achieved relations and control events.

3.2 Unified Agent Harness

Model-directed interaction. The harness supports persistent model-directed interaction (Figure 2d). Geometry, Spatial Reasoning, Affordance, and Planning use a Python runtime; Manipulation follows its native agent loop. Models interpret observations, select operations, and construct action sequences using the resulting feedback. The main evaluation tests these model-directed decisions through source-native helpers; specialist measurement tools are introduced only in the supplementary review study. Variables persist across Python rounds, and images produced during a round enter the next model request. Appendix B.1 specifies these interfaces; Section 4.4 examines multi-round review with specialist visual tools (Zhang et al., 2025, Huang et al., 2026c).

Submission and termination. Fixed-input submissions commit predictions; interactive runs end on agent termination, an environment terminal condition, or budget exhaustion. Scoring independently checks outputs or states. Appendix B.2 details budgets.

3.3 Evaluation Protocol and Metrics

① *Task success.* Spatial Reasoning and Planning report Pass@1. Manipulation reports native success, consecutive goals completed, and task-specific condition or control-evidence scores. Categorical spatial accuracy is reported separately. ② *Geometric precision.* Geometry and numerical spatial queries use MAE, RMSE, AbsRel/MAPE, displacement error, and pose errors according to the target. Tracing pairs trajectory RMSE with offline 3D constraint satisfaction. ③ *Native affordance grounding.* Predicted boxes use GT-mask IoU, with contact validity scored separately. Box–box IoU and strict box success at IoU≥ 0.5 measure localization using the same rectangular representation. ④ *Variation and resource use.* Repeated runs report the mean and standard deviation (SD) across runs; SD measures run-to-run variation. Tokens measure inference use; action calls measure executed control. Section 4.1 specifies repetitions; Appendix B.3 defines aggregation.

4 Experiments

4.1 Experimental Setup

① *Models and configuration.* We compare six closed-source VLMs (Astra, Sol, Fable, Opus, Qwen-Max, Gemini) and two open-source VLMs (Qwen-397B, Qwen-27B). Appendix B.2 records identifiers and source-specific limits. The eight-agent result snapshot is distinguished from the original seven-agent protocol diagnostics. ② *Evaluation and aggregation.* Original fixed-input tasks use five runs per case, or three for Fable. Opus uses five runs on non-tracing Geometry and one elsewhere. Continuous metrics pool valid observations; rates retain assigned cases. Opus Planning excludes HumanCLAW while its results are reconciled, so it covers 144 cases rather than 157. Manipulation retains all 182 assignments, counting two Opus RLBench errors and one RoboTwin error as unsuccessful. All eight agents use the same Baseline Manipulation protocol; the original seven scores remain unchanged. Appendix C.3 details Opus coverage and repetition accounting.

4.2 Main Results

Local strengths differ across models and metrics. Astra has the lowest pooled pose errors (17.1°, 106.6 cm) and highest full-allocation Planning success (80.3%). Its Spatial Pass@1 narrowly exceeds Qwen-Max's (69.4% versus 68.9%), whereas Gemini has lower numerical AbsRel (0.161 versus 0.213). Opus reaches mask IoU 0.329 and contact validity 62.9%, above Astra's 0.286 and 60.3%. Opus has one Affordance run, so this point-estimate lead is not a repeated-run significance claim. Figure 3 shows the resulting metric profiles.

Local accuracy does not establish complete execution. Opus has the highest observed Baseline Manipulation success at 47.8%, versus Astra's 41.8%. Both use one episode per assignment, so this six-point difference describes the evaluated task set rather than repeated-run reliability. The original matched-budget Baseline/RGB-expanded experiment remains in Section 4.4. Across the supplied household records, all eight models have zero complete success

Table 2. Performance across five capability domains. ± denotes run SD; continuous means use valid observations. Opus uses five non-tracing Geometry runs and one elsewhere. †Opus Planning: 73/144, excluding HumanCLAW; other agents: 157 cases. Manipulation uses Baseline for all eight agents (182 assignments); Opus achieves 87 successes. Bold marks the best comparable mean.

	Geometry		Spatial Reasoning		Affordance		Task Planning	Manipulation
Agent	Rot.↓ (°)	Trans.↓ (cm)	Pass@1↑ (%)	AbsRel↓	Mask IoU↑	Point↑ (%)	Pass@1↑ (%)	Task SR↑ (%)
Astra	**17.1**	**106.6**	**69.4**±1.0	0.213	0.286	60.3±2.6	**80.3**	41.8
Sol	27.0	151.9	60.7±2.5	0.273	0.264	52.0±3.6	44.6	19.8
Fable	25.2	121.7	62.0±4.6	0.186	0.284	57.1±2.5	63.7	24.7
Qwen-Max	20.5	138.1	68.9±1.1	0.205	0.272	52.9±4.2	35.7	9.3
Gemini	24.2	140.9	65.4±1.3	**0.161**	0.250	44.6±13.0	48.4	18.7
Qwen-397B	38.4	249.8	50.6±1.8	0.251	0.116	32.6±4.7	17.8	9.9
Qwen-27B	38.4	213.7	48.0±0.9	0.288	0.013	6.3±1.6	19.1	11.0
Opus	23.3	135.3	58.6	0.222	**0.329**	**62.9**	50.7†	**47.8**

on five BEHAVIOR-1K tasks. This means the native final conditions were not all satisfied under the respective execution limits, not that every episode timed out.

4.3 A Deep Dive of Astra

Figures 4–8 retain the original seven-agent subtask and protocol diagnostics; their peers exclude the later Opus evaluation. Appendix D lists those task groups, while Table 2 gives the updated eight-agent summary and Table A11 gives Opus source results. Within the original cohort, Astra leads Spatial, contact localization, and Planning under equal source weights (Appendix C.2).

Geometry. ① *Astra leads controlled metric estimation across targets.* It has the lowest errors on all five Blender estimation targets (Figure 4a), including reference height at 0.31% versus Sol's 0.90%. Depth ordering reaches 96.3%, while oracle scale alignment lowers depth AbsRel from 0.104 to 0.038: much of its remaining depth error concerns scale rather than order. InFlux focal error reverses the ranking (53.0% versus Sol's 47.8%), locating the strongest advantage in controlled geometric cues. ② *Astra separates object displacement from moderate viewpoint change.* Across fixed, small-shift, and large-shift Blender conditions, Astra's MAE is 0.24, 0.57, and 1.13 cm; Sol is the closest peer throughout (Figure 4b). The fixed-camera lead survives excluding stationary objects. On four real tabletop photo pairs, Astra achieves 0.05 versus Sol's 0.42 cm MAE. The lower real-scene error occurs on different scenes and object motions. ③ *Coupled motion challenges Astra's otherwise leading pose estimates.* Its aggregate pose errors are lowest (Table 2), yet no model meets both rotation and translation tolerances under mixed motion. ④ *Astra traces the closest paths, but endpoints decide success.* Astra has the lowest 3D trajectory RMSE at 0.382 m versus Gemini's 0.426, whereas joint success is 33.9% versus 37.5% (Figure 4c). Initiation or arrival fails in 36 of Astra's 37 unsuccessful traces, identifying endpoint placement as the main obstacle to successful tracing.

Spatial Reasoning. ① *Astra's clearest advantage is relating multiple views.* Across all MMSI tasks it reaches 77.0% versus Fable's 66.7%; across MindCube it ties Qwen-Max at 90.8%. Within these sources, object relations reach 96.7% and camera–region relations 100% (Figure 5a,c). Equal weighting of all six sources also ranks Astra first, at 71.9% versus Qwen-Max's 69.1% (Appendix C.2). ② *Astra's strength in relations outpaces its inference of new frames.* MindCube camera-shift accuracy falls to 78.5% versus Qwen-Max's 89.2%; 3DSRBench object-facing judgments reach 56.7% versus Fable's 72.2% (Figure 5a). The shared-run gaps in Figure 5c preserve both deficits. Relating visible landmarks and recovering motion or an object's own axes make

Figure 3. Eight-agent fixed-input metric profiles. Each panel retains its native units and direction. Dots show reported means, not confidence intervals; Opus has one run outside non-tracing Geometry. Interactive task success remains in Table 2; its Opus Planning entry excludes HumanCLAW.

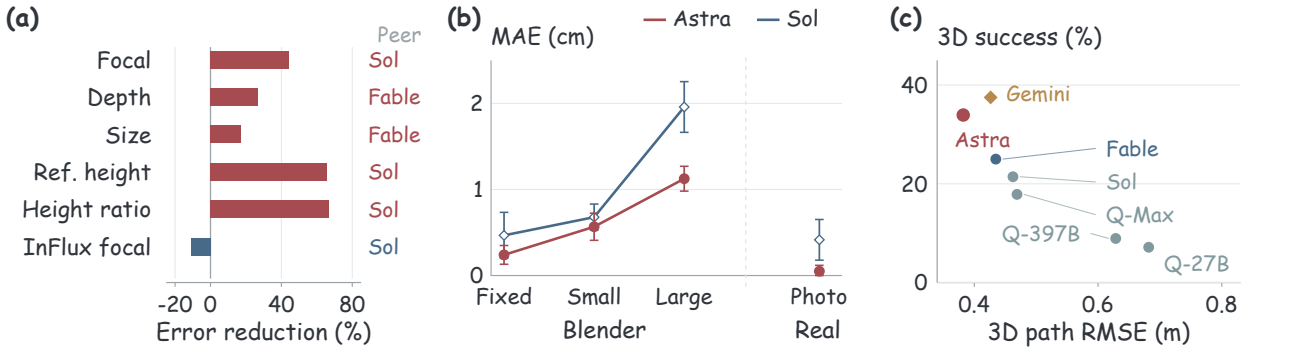

Figure 4. Original seven-agent diagnostic. Geometric precision and task completion. (a) Error reduction relative to each target's lowest-error peer; the first five targets are GeoProbe. (b) Displacement under fixed, small-shift, and large-shift Blender cameras, plus separate real photos; points show mean ± run SD. (c) Single-run 3D tracing error versus joint success. Appendix D.1 gives absolute metrics and scene definitions.

different demands. ③ *Depth cues are more reliable than choosing among objects.* Astra achieves 96–100% on BOP-ASK camera-depth ordering, but 50% on VSI nearest-object selection versus Qwen-Max's 76.7% (Figure 5b). RoboSpatial configuration is also 22 points behind its best peer (Figure 5c). Thus its narrow overall Spatial lead combines strong multi-view relations with unresolved object-centered selection and configuration.

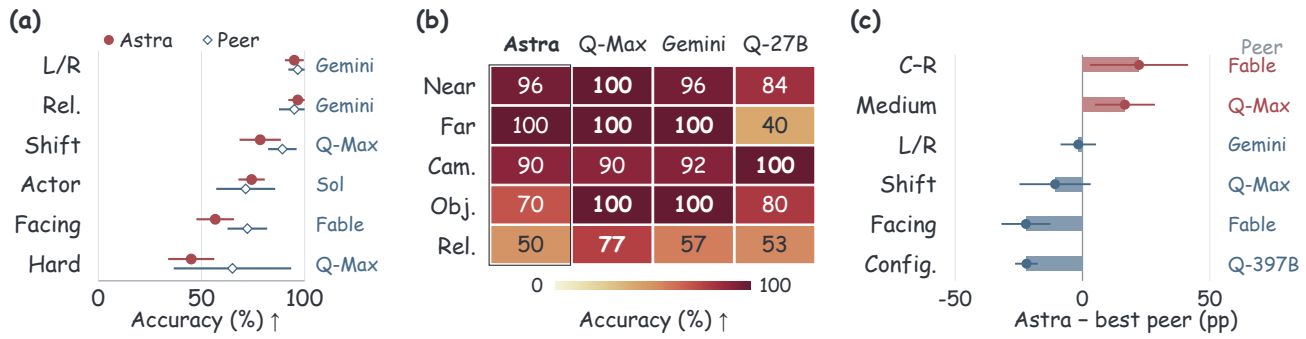

Figure 5. Original seven-agent diagnostic. Astra's spatial strengths and reference-frame limits. (a) Relation and viewpoint accuracy compared with each task's best peer. (b) Camera-relative depth (Near/Far/Cam.) and inter-object distance (Obj./Rel.). (c) Astra's accuracy advantage over the best peer on one task per source; C–R denotes camera–region relations. Panels a–b average available runs; c compares shared run indices. Whiskers show run-to-run SD. Appendix D.2 maps labels to all six sources and reports every task.

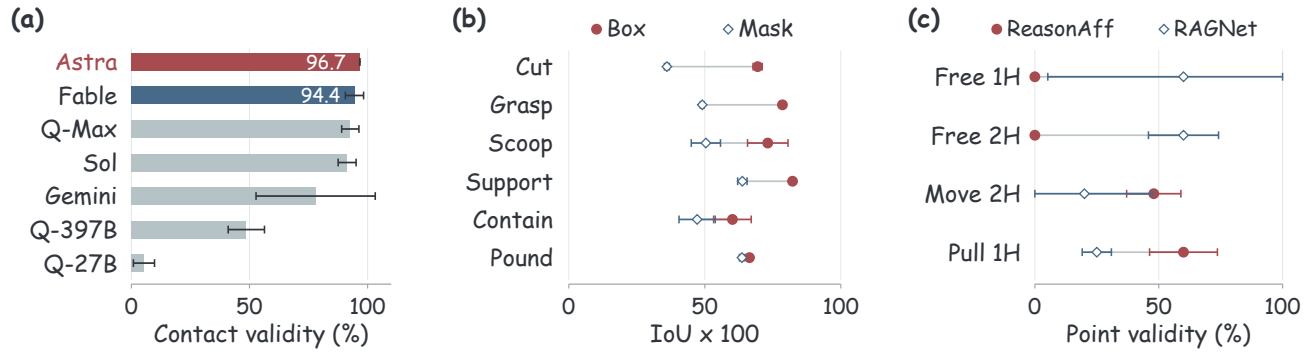

Figure 6. Original seven-agent diagnostic. Astra's native affordance localization. (a) Contact validity over all six UMD actions. (b) Astra's boxes evaluated against reference boxes and masks. (c) Contact validity for four corresponding interactions in ReasonAff/RAGNet; 1H/2H means one/two hands. Scores show mean ± run SD; IoUs are multiplied by 100. Appendix D.3 gives all actions and models.

Affordance. ① *Astra's contact-localization lead extends across sources.* It leads UMD and ReasonAff at 96.7% and 31.0%, versus Fable's 94.4% and 23.3%, and ties Fable at 35.0% on RAGNet. Figure 6a shows the UMD ranking across all six actions. Across these sources, absolute contact accuracy varies more than Astra's relative ranking. ② *Native boxes locate usable parts without reproducing their shape.* Cutting reaches 100% contact validity and 96% strict box success, despite box–mask IoU of 0.361; box–box IoU is 0.694 (Figure 6b). The rectangular output explains part of this metric gap. Containment still has only 44% strict box success versus Qwen-Max's 96%, showing that contact localization and region extent are distinct even under the same output format. ③ *Action-conditioned contacts expose where Astra's lead changes.* ReasonAff pulling and two-handed movement reach 60% and 48%, above the best peers' 50% and 26.7%; the RAGNet counterparts reach 25% and 20%, below 31.7% and 50% (Figure 6c). The same panel shows zero success on both ReasonAff free-movement categories despite nonzero RAGNet scores. Contact choice therefore depends on the action–scene pairing, beyond recognizing a tool's functional part.

Task Planning. ① *Astra's planning lead spans all six sources.* It leads four sources and ties the best model on two (Figure 7a; Appendix D.4). ALFRED and ALFWorld success is 75% and 70% versus Fable's 35% and 30%. Astra alone solves ALFRED heating and dual placement (Figure 7b), extending its advantage to state changes and maintaining progress across multiple objects. ② *Astra gains most when scientific goals require linked operations.* DiscoveryWorld

reaches 100% versus Fable's 70%; ScienceWorld reaches 87.7% versus 80.8%. Figure 7c locates the largest gains in unknown-plant genetics and melting-point measurement. Thermometer use reverses this lead at 66.7% versus 83.3%, identifying instrument operation as a distinct bottleneck within scientific planning. ③ *Astra finds seating targets but fails to establish the sitting state.* It ties Gemini for the best HumanCLAW contact rate (60%), yet all seven original agents fail sitting. Five Astra episodes reach the seat and issue the action, whereas Fable records no target-finding success in these episodes. This execution evidence distinguishes locating a target from establishing the required body–object state.

Manipulation. ① *Astra's baseline lead spans several operation families.* It leads eight of thirteen suites among available baseline records (Figure 8a). Examples include CLIPort at 61.1% versus Fable's 33.3%, RLBench at 60% versus Sol's 20%, and RoboTwin at 38% versus Fable's 14%, spanning placement, articulation, and coordinated handling. Its five-of-nine ManiSkill success is supported by a 60% condition score on the diagnostic subset (Figure 8c). ② *Astra converts baseline control into longer goal sequences.* Astra completes 3.10 consecutive CALVIN goals on average versus Gemini's 1.80, Sol's 1.40, and Fable's 0.90 (Figure 8b). Only Astra finishes full sequences, on three of ten tasks. The expanded-condition markers in panels b–c test the setting dependence of this advantage in Section 4.4. ③ *Household execution exceeds Astra's local control strengths.* Within the source-specific execution budgets, Astra completes one of ten RoboCasa tasks and none in RoboCasa365; all seven original agents fail the five BEHAVIOR-1K tasks (Figure 8a). These tasks combine mobile-base positioning and articulated fixtures with object handling (Nasiriany et al., 2024). Household success requires local control to support coordinated navigation, placement, and environment-state changes.

4.4 Further Investigations

Richer observations under matched budgets. These controlled comparisons retain the original seven-agent manifests; the updated snapshot is reported separately. ① *Astra's control lead can reverse.* With inference budgets and execution-step limits fixed, per-round RGB changes CALVIN success from 30% to zero for Astra and zero to 40% for Fable (Table 3b); average consecutive-goal scores also reverse (Figure 8b). ② *Astra processes more input while executing fewer actions.* Astra's median input tokens rise from 0.32M to 1.21M, while action calls fall from 1,795 to 481. In sequence 0557, its reusable controller appears at round 12 rather than 5; expanded Fable builds one at round 2. This trajectory contrast links completion to when agents establish reusable feedback control (Appendix E.1). ③ *Stable totals hide opposing task effects.* VIMA improves for all three models, and RoboTwin improves for Astra. Across 153 common tasks, Astra stays at 63 successes while Fable rises from 34 to 65. ManiSkill condition scores likewise fall from 60% to zero for Astra, while expanded Fable reaches 60% (Figure 8c). Task-specific gains offset losses in Astra's aggregate.

Multi-round review on fixed inputs. Agents revisit initial answers on 46 shared cases. Self-review and Crop review allow two additional responses; Tool-assisted review combines six additional responses with specialist measurements (Appendix E.2). Tool-assisted review raises Astra's Affordance success from 22.2% to 33.3%, but improves five and worsens four Geometry entries (Table 3a). Astra's Spatial accuracy remains at 72.2%, while Qwen-27B rises from 55.6% to 61.1%. Qwen-27B's self-review yields higher box IoU than tool assistance (0.1524 versus 0.1452), yet tool assistance alone produces a box passing the 0.5 threshold. Mean overlap and threshold success capture different effects of review. The tool-assisted condition measures the joint effect of added measurements and a larger inference budget.

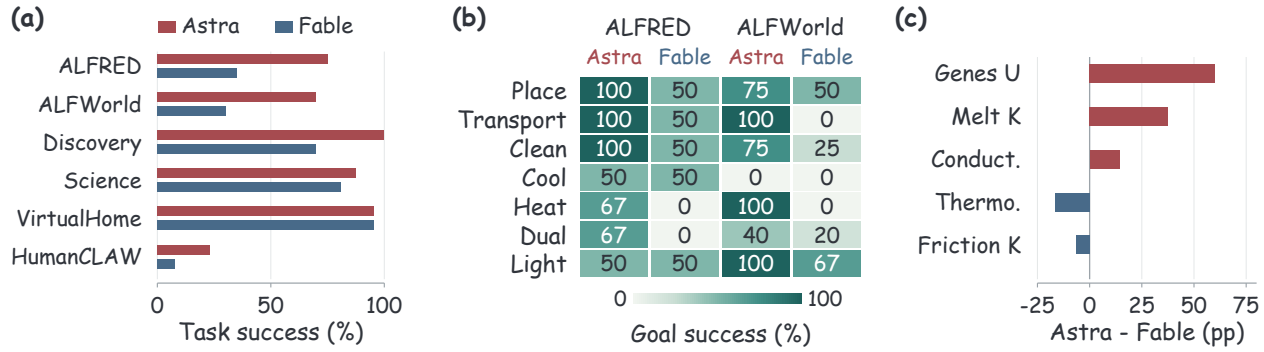

Figure 7. Original seven-agent diagnostic. Astra's planning advantage across sources and goals. (a) Overall success for all six sources. (b) ALFRED/ALFWorld success on seven household goal families. (c) All five nonzero Astra–Fable gaps in ScienceWorld; thirteen families tie. K/U means known/unknown conditions. Single-run task outcomes; Appendix D.4 includes all seven models.

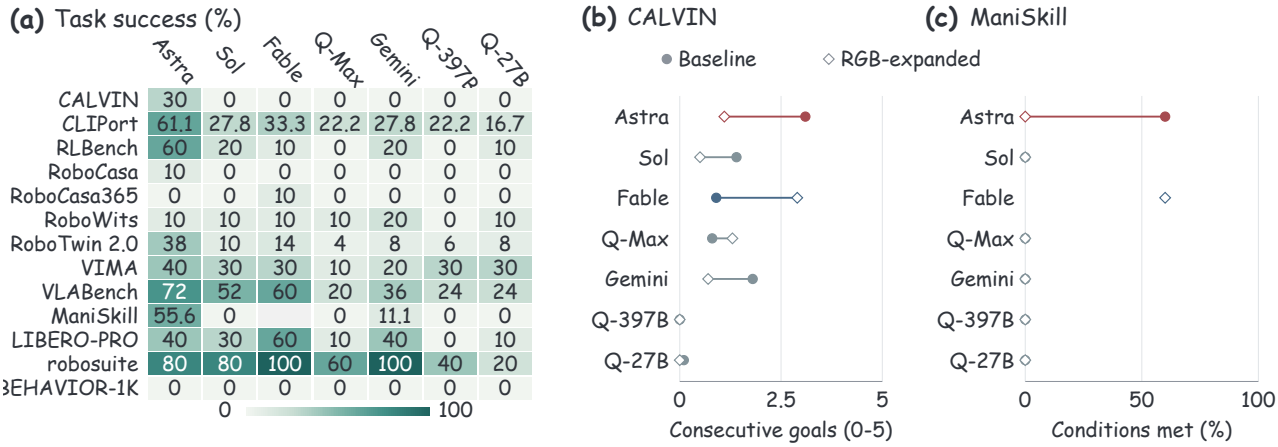

Figure 8. Original seven-agent diagnostic. Manipulation across suites and execution settings. (a) Task success on thirteen suites; the last three rows use the same records in both protocols. (b) Average CALVIN goals completed consecutively before the first failure, out of five. (c) ManiSkill goal-condition satisfaction on five diagnostic tasks. Filled/open markers show Baseline/RGB-expanded settings (Section 4.4). Each task uses one run; Appendix D.5 gives metrics and coverage.

Case studies. Appendix F links spatial judgments and contact localization to concrete inputs, then compares controller construction and goal completion in CALVIN and PickCube. CloseLaptopLid and PullCube frames contrast visible motion with complete execution.

Together, these cases connect local model decisions to the conditions left unsatisfied at termination. Appendix E.3 relates these outcomes to task difficulty and external execution support.

5 Conclusion

Embodied Agent Arena evaluates native VLM perception, reasoning, and action selection across robotic tasks under source-specific interfaces. Across 1,000 assigned cases and eight agents, Astra leads pose estimation and planning, whereas Opus achieves the highest measured affordance overlap, contact validity, and Baseline manipulation success. Accurate local predictions still coexist with boundary errors and incomplete household goals. Richer observations can also reverse control rankings. The bottleneck is precise completion within finite budgets: repeated decisions consume time and tokens, while endpoint and final-state errors leave goals unsatisfied.

Table 3. Multi-round review (a) and RGB-expanded Manipulation (b). Baseline and RGB-expanded share budgets and execution-step limits; RGB-expanded supplies per-round images. I/T/W counts improved/tied/worsened entries; SR_{50} requires box IoU≥ 0.5. Goals gives average consecutive completions (0–5); Common pools nine suites (153 tasks). Details: Appendix E.

(a)		Geometry	Spatial Reasoning	Affordance	
Model	Condition	I/T/W	Acc. (%)↑	Box IoU↑	SR_{50} (%)↑
Astra	Initial	–	**72.2**	0.2903	22.2
	Self-review	3/20/1	66.7	0.2914	22.2
	Crop review	4/19/1	66.7	0.2912	22.2
	Tool-assisted	5/15/4	**72.2**	**0.3174**	**33.3**
Qwen-27B	Initial	–	55.6	0.0336	0.0
	Self-review	6/15/3	50.0	**0.1524**	0.0
	Crop review	3/19/1	44.4	0.0937	0.0
	Tool-assisted	10/6/8	**61.1**	0.1452	**11.1**

(b)		CALVIN		ManiSkill	VIMA	Common
Model	Condition	SR (%)↑	Goals↑	SR (%)↑	SR (%)↑	SR (%)↑
Astra	Baseline	**30.0**	**3.10**	**55.6**	40.0	**41.2**
	RGB-expanded	0.0	1.10	11.1	**100.0**	**41.2**
Sol	Baseline	0.0	**1.40**	**0.0**	30.0	19.0
	RGB-expanded	**10.0**	0.50	**0.0**	**90.0**	**23.5**
Fable	Baseline	0.0	0.90	–	30.0	22.2
	RGB-expanded	**40.0**	**2.90**	55.6	**100.0**	**42.5**

The arena covers bounded tasks with source-specific helpers and budgets. Interactive results use one episode per assignment; Opus HumanCLAW remains excluded pending verification. Public task assets leave unseen-environment generalization open. More repetitions, longer tasks, and physical-robot deployments would test whether agents can preserve achieved conditions across successive actions while satisfying complete goals.

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Appendix: Supporting Evidence and Protocols

Reading guide. Appendices A–B define task construction, interfaces, and scoring. Appendix C expands Table 2, including source-composition sensitivity and Opus dispersion. Appendix D follows the five domains of Section 4.3, from full task comparisons to Manipulation's auxiliary metric definitions. Appendix E specifies the execution-protocol and multi-round review comparisons, and Appendix F illustrates recorded actions and terminal outcomes, with additional examples across all five capabilities in Appendix H. Appendix G gives the evaluation scales and public resources behind Table 1.

A Benchmark Construction and Source Inventory

Embodied Agent Arena contains 1,000 cases from 32 established sources and our GeoProbe benchmark (Table A1). A case is one query or episode; multiple queries can share a scene. Benchmark-specific allocation uses seed 42 and targets diagnostic coverage of the five capabilities in Section 3.1.

Table A1. The unified 1,000-case arena, drawn from 32 established sources and our GeoProbe benchmark. Detailed source allocations follow below.

Domain	Cases	Share (%)	Task coverage
Geometry	370	37.0	Calibration, pose, scale, tracing
Spatial Reasoning	220	22.0	Relations, views, video
Affordance	70	7.0	Functional parts, contact
Task Planning	157	15.7	Household, navigation, science
Manipulation	183	18.3	Robot operations, sequences
Total	1,000	100.0	660 fixed-input; 340 interactive

Query vocabulary. Figure 2a counts each term once per query, balances weights across source groups and domains, and includes English translations of GeoProbe queries. GeoProbe's ten protocols form one source group.

Geometry (370 cases). InFlux-Real (Liang et al., 2025) contributes 30 single-frame intrinsic-estimation cases, of which four have invalid reference annotations and are excluded from continuous-error calculations. Map-free (Arnold et al., 2022) contributes 30 relative-pose image pairs with known camera intrinsics. MultiSPA-derived uses Map-free images and poses with the camera-motion question formulation associated with Multi-SpatialMLLM (Xu et al., 2025): 12 angle, ten distance, and eight displacement-vector questions. This local adaptation retains the Map-free image and pose references. The pose transformation and camera-center displacement use distinct coordinate conventions and reference targets.

GeoProbe comprises ten diagnostic protocols: 168 queries over 130 unique RGB images, predominantly rendered in Blender, with four real-photo displacement cases. Shared scenes separate calibration, pose, depth, displacement, and scale requirements (Table A2).

TraceSpatial-Bench (Zhou et al., 2026), introduced with RoboTracer, adds paired 2D and 3D queries on 56 scenes. These 112 Geometry cases predict offline object trajectories, scored for reference agreement and geometric constraints.

Table A2. GeoProbe's ten diagnostic protocols, with shared images and scenes.

Diagnostic	Cases	Diagnostic	Cases
Object displacement, fixed camera	12	Full camera intrinsics	24
Object displacement, small view change	12	Focal length, centered principal point	12
Object displacement, large view change	12	Relative camera pose	24
Real tabletop displacement	4	Sparse depth	12
Object dimensions	12	Reference-based scale	44

Spatial Reasoning (220 cases). 3DSRBench (Ma et al., 2024) contributes 50 relation questions; 28 flip-tagged entries were displayed unflipped. Scores use corrected references for five affected answers under the unflipped presentation. MindCube (Wang et al., 2025b) contributes 50 multi-view cases: 39 among-view, six around-view, and five rotation. VSI-Bench (Yang et al., 2024) contributes 50 video questions: nine absolute-distance, nine object-size, fourteen relative-direction, six relative-distance, five counting, four room-size, two appearance-order, and one route-planning. Eight frames form its initial observation. RoboSpatial-Home (Song et al., 2024) contributes ten cases each for compatibility, configuration, and context. BOP-ASK (Bhat et al., 2025) contributes twelve relative-position and eight relative-depth questions. MMSI-Bench (Yang et al., 2025b) contributes 20 cross-image questions. Together these comprise 183 categorical, 27 numerical, and ten point-set questions, scored as defined in Appendix B.3.

Affordance (70 cases). UMD (Myers et al., 2015) contributes 30 views spanning five tool classes and six functions: grasp, cut, scoop, contain, pound, and support. Multiple views can depict the same object. The 20 ReasonAff-style cases adapt instruction-conditioned queries motivated by ReasonAff (Wang et al., 2025a) to 3DOI (Qian & Fouhey, 2023) interaction annotations. Another 20 RAGNet-3DOI cases use explicit action-conditioned queries (Wu et al., 2025). The two subsets use disjoint images. These images and annotations support native point and region evaluation.

Task Planning (157 cases). ALFRED (Shridhar et al., 2019) and ALFWorld (Shridhar et al., 2020) contribute 20 visual cases each, covering placement, heating, cooling, cleaning, containers, multiple objects, and illumination. ScienceWorld (Wang et al., 2022) contributes 73 text-based cases across eighteen topics. VirtualHome (Puig et al., 2018) contributes 21 symbolic programs in seven scenes. The composition analysis removes these 94 cases together. DiscoveryWorld (Jansen et al., 2024) contributes ten tasks with structured observations. HumanCLAW contributes thirteen Habitat navigation and sitting episodes: five sofa, three toilet, two chair, two plant, and one television target. Observations are visual, textual, or symbolic.

Table A3. Manipulation tasks across 13 sources, with identical assignments for all agents.

Benchmark	Cases	Task coverage
CALVIN (Mees et al., 2021)	10	Ten sequences, each with five ordered subgoals
CLIPort (Shridhar et al., 2021)	18	Eighteen task and color-generalization configurations
RLBench (James et al., 2019)	10	Ten task configurations
RoboCasa (Nasiriany et al., 2024)	10	Seven atomic and three composite tasks
RoboCasa365	10	Three atomic and seven composite tasks
RoboWits	10	Ten official task configurations
RoboTwin 2.0 (Chen et al., 2025a)	50	Fifty distinct manipulation tasks
VIMA-Bench (Jiang et al., 2023)	10	Six placement-generalization and four new-task cases
VLABench (Zhang et al., 2024)	25	8 ID, 3 category, 4 commonsense, 7 semantic, 3 texture
ManiSkill (Tao et al., 2024)	10	Nine binary tabletop tasks; one quadruped reward task
LIBERO-PRO (Fu et al., 2026)	10	Ten object, spatial, goal, and multi-object configurations
robosuite	5	Two lift seeds; stacking, restacking, and wiping
BEHAVIOR-1K (Li et al., 2024a)	5	Five multi-object household activities

Manipulation (183 cases). Table A3 lists 182 binary tasks and one continuous-reward task. LIBERO-PRO, robosuite, and BEHAVIOR-1K use benchmark-specific code interfaces (Fu et al., 2026). RoboCasa365 uses variations designated for pretraining in the source inventory. Appendix B.1 specifies the native assistance available to agents.

B Evaluation Protocol and Reproducibility

B.1 Observations and Model-Directed Interaction

Geometry, Spatial Reasoning, and Affordance inspect fixed observations; Planning and Manipulation interact with an environment. All domains except Manipulation use a persistent Python runtime; Manipulation uses its native agent loop. Both enforce episode-level budgets and termination. Variables and environment state persist between code rounds, and images produced by a cell reach the next model request after that cell ends.

Table A4. Model-visible observations, actions, and feedback. Private scoring annotations are unavailable to the agent.

Domain	Observation support	Actions and feedback
Geometry	Image or image pair; intrinsics for Map-free	One final numerical submission
Spatial Reasoning	Images, crops, and video-frame queries	Evidence inspection and answer submission
Affordance	Image and interaction instruction	One point or bounding-region submission
Task Planning	Images, text, or symbolic state	Navigation, interaction, programs, public progress
Manipulation	Protocol-specific scene/state observations and native grounding helpers	Motion or skill calls; execution and task checks

ALFRED and visual ALFWorld expose observation, search, navigation, interaction, and memory primitives. CLIPort and VIMA expose instance inspection and pick-and-place actions.

RoboTwin provides actor localization, pose inspection, planning probes, and execution. LIBERO-PRO, robosuite, and BEHAVIOR-1K expose benchmark-specific observation and action interfaces; RoboCasa includes fixture and contact localization. The model makes decisions using these native helpers; the baseline adds no specialist perception or verification pipeline. Geometry requests direct visual estimates and excludes pose-solver implementation.

A runtime exception can be followed by another code round if budget remains; prior state changes are not automatically undone. Model requests, code rounds, primitive calls, and environment steps count different units. Stored visualization frames are also distinct from images actually sent to the model. Qualitative figures sample the recorded camera streams.

B.2 Models and Execution Budgets

Table A5. Models and configured identifiers for all eight agents in the five domains.

Model label	Identifier
GPT-6 Astra	`gpt-6-astra`
GPT-6 Sol	`gpt-6-sol`
Claude Fable 5.1	`claude-fable-5-1`
Claude Opus 5.5	`claude-opus-5-5`
Gemini 3.8 Flash	`gemini-3.8-flash`
Qwen 3.8 Max	`qwen3.8-max`
Qwen 3.5 397B-A17B	`qwen3.5-397b-a17b`
Qwen 3.5 27B	`qwen3.5-27b`

The original fixed-input and Planning configurations use temperature zero. Opus uses the recorded identifier above and the same Baseline Manipulation protocol as the original seven agents. Baseline request limits are 1,800 output tokens and 360 seconds; fixed-input profiles also allow up to 16,384 tokens and 900 seconds. On the configured CLI interface, reasoning effort is maximum for Sol and high for Astra and Fable; API requests omit this parameter. Astra's first repetition on the original 90-case Geometry core uses the high-effort CLI configuration. Manipulation's request settings and shared execution limits are specified in Appendix E.1.

Table A6. Baseline episode budgets. Tokens accumulate within each case; rounds are code interactions, not repetitions. Dashes denote unspecified fields. Appendix B.4 lists fixed-input settings.

Entry	Code rounds	Tokens	Env. steps	Primitives	Episode seconds	Task seconds
Geometry	12	120k	–	–	–	960
Spatial: non-VSI	6	96k	–	–	900	960
Spatial: VSI	8	240k	–	–	1,200	1,500
ALFRED, ALFWorld	8	320k	120	540	1,200	1,500
DiscoveryWorld	8	320k	100	400	1,200	1,500
HumanCLAW	101	2,000k	100	400	10,800	11,000
ScienceWorld	8	320k	100	500	1,200	1,500
VirtualHome	8	320k	16	120	1,200	1,500
Affordance	12	120k	–	–	–	960

Execution and native-call time limits are clipped by remaining budgets. Appendix E.1 gives Manipulation's shared execution limits, camera presets, and native step-limit semantics.

B.3 Scoring and Repeated Evaluation

Submission and termination. A final submission commits a fixed-input prediction. Interactive runs end when the agent stops, the environment reaches a terminal condition, or a source-specific round, token, action, or time limit is reached. The evaluator scores the submitted output or recorded state independently of the agent's declaration. A complete episode includes all its internal reasoning, actions, and recovery; these steps are not separate attempts.

Task scores and aggregation. Categorical questions use answer accuracy and interactive tasks use native goal success. Invalid or absent predictions remain in success denominators. Invalid reference annotations are excluded before scoring. Continuous errors pool valid predictions with compatible metrics and units, alongside their coverage. Numerical Spatial Reasoning uses exact-match success, $|\hat{y} - y| < 10^{-6}$, alongside continuous errors; this differs from VSI-Bench's official mean relative accuracy. Point-set questions retain their local set criterion.

Pass@1 and repeated scores. For N assigned cases and R repetitions, let $s_{ir} \in \{0, 1\}$ denote success on case i in run r. The average single-episode success rate is

$$\text{Pass@1} = \frac{1}{R} \sum_{r=1}^{R} \left(\frac{1}{N} \sum_{i=1}^{N} s_{ir} \right). \tag{2}$$

This estimates success in one complete episode under the specified condition. Fresh agent state is used for formal repetitions. For any per-run score M_r, we compute

$$\bar{M} = \frac{1}{R} \sum_{r=1}^{R} M_r, \qquad v_M = \frac{1}{R-1} \sum_{r=1}^{R} (M_r - \bar{M})^2, \qquad \text{SD}_M = \sqrt{v_M}. \tag{3}$$

Tables using $\bar{M} \pm \text{SD}_M$ report variation between run-level scores. Variance is supplied separately in squared metric units. Figure 4 averages valid-case errors within each run and then weights the recorded runs equally. Its coverage sensitivity instead fixes the cases valid in every shared repetition for each model pair; those common-cohort estimates are reported separately. Figure 5 uses per-run categorical accuracies; its pairwise gaps average accuracy differences over shared run indices, with the comparator fixed per subtask. Run indices align reporting repetitions; seeds and tool configurations can differ. Run-to-run SD requires $R > 1$.

Scalar estimates and depth. For J valid targets, $e_j = \hat{y}_j - y_j$ gives $\text{MAE} = J^{-1} \sum_j |e_j|$ and $\text{RMSE} = \sqrt{J^{-1} \sum_j e_j^2}$. Absolute relative error is $\text{AbsRel} = J^{-1} \sum_j |e_j|/|y_j|$ on nonzero targets, and $\text{MAPE} = 100\,\text{AbsRel}$ is expressed in percent. Size and focal-length estimates use absolute and relative errors. Depth additionally reports RMSE in meters. Depth errors are computed over positive finite predictions within each image, then averaged across images, with valid-point coverage reported. The fraction within 10% relative error and $\delta_1 = J_{\text{all}}^{-1} \sum_j \mathbf{1}[\max(\hat{y}_j/y_j, y_j/\hat{y}_j) < 1.25]$ use all queried points as their denominator; invalid predictions contribute zero. Ordinal accuracy is the fraction of eligible point pairs with the correct depth ordering, with invalid predictions counted as incorrect. Scale alignment removes the global depth factor.

Displacement and camera pose. For a nonzero scalar or vector target y, motion relative error is $\|\hat{y} - y\|_2 / \|y\|_2$. The MultiSPA-derived tolerance is 20%; a zero target requires zero absolute error. GeoProbe object-displacement diagnostics separately test movement status, direction, absolute distance error, and direction agreement with distance error at most 1 cm. Stationary cases are evaluated without division by zero. Camera translation error is $e_t = \|\hat{\mathbf{t}} - \mathbf{t}\|_2$, reported in centimeters. Rotation error is $e_R = \arccos[(\text{tr}(R^\mathsf{T} \hat{R}) - 1)/2]$, in degrees, clipping the argument to $[-1, 1]$. GeoProbe reports component errors and joint success within 2° and 2 cm.

Camera calibration. Focal MAPE averages the absolute relative errors of f_x and f_y per case. Principal-point error is $e_c = \|\hat{\mathbf{c}} - \mathbf{c}\|_2$ in pixels, also normalized by the image diagonal D. Joint calibration requires both focal components to satisfy 10% relative error and $e_c \leq 0.02D$. Main-table pose errors and complementary continuous metrics use pooled valid-observation means; Figure 4 instead uses equal-run means, as specified above.

Spatial tracing. For each case, predicted and reference paths are independently resampled to $L = 100$ equally spaced arc-length positions. Trajectory error is $\mathrm{RMSE}_{\mathrm{trace}} = \sqrt{L^{-1}\sum_{\ell=1}^{L}\|\hat{\mathbf{p}}_\ell - \mathbf{p}_\ell\|_2^2}$. We average this per-case value over valid scenes. The 2D metric uses normalized image coordinates; the 3D metric uses meters. The two metrics retain their respective units. The supplied variance is the population variance of the 56 per-scene RMSE values for a single run.

Offline 3D success requires valid initiation, target arrival, and collision checks to pass together. The predicted start must lie within 20 cm of the object point cloud or the reference start. At least one of the final three predicted points must lie within 20 cm of the target 3D box; for two-point paths, only the endpoint is checked. Translating the object point cloud to each checked predicted position must yield at most 20% overlap with the environment. Collision is checked at the sampled positions. Missing or invalid trajectories count as failures. 2D tracing reports continuous error. Qwen-27B trajectory outputs undergo format normalization that preserves all predicted coordinate values.

Native affordance grounding. For a directly predicted bounding box B and ground-truth mask G, the box interior is rasterized and $\mathrm{IoU}_{\mathrm{mask}} = |B \cap G|/|B \cup G|$. Contact validity is $\mathbf{1}[p \in G]$ for the submitted point p, with a reference-box fallback only where a mask is unavailable. A separate diagnostic uses $\mathrm{IoU}_{\mathrm{box}} = |B \cap B_G|/|B \cup B_G|$ against the reference box B_G. Strict box success requires $\mathrm{IoU}_{\mathrm{box}} \geq 0.5$; contact validity is scored independently. The source field named `giou` contains ordinary box–mask IoU, not generalized IoU.

Resource accounting. Inference tokens, action calls, model rounds, and elapsed time measure different aspects of execution. The Manipulation comparison reports token usage and action calls from the supplied execution analysis; protocol-matched per-case durations are unavailable. Request, trial, and native-call time limits are specified separately in Appendix E.1.

B.4 Repetition Protocol and Result Accounting

Repetitions. In the original seven-agent cohort, non-tracing Geometry, Spatial Reasoning, and Affordance use five repetitions per case for six models and three for Fable. Opus uses five runs for non-tracing Geometry and one for tracing, Spatial Reasoning, Affordance, Planning, and Manipulation; no run-to-run SD is inferred for its single-run results. Spatial tracing, Planning, and Manipulation have one reported episode per assignment and therefore no run-to-run SD. Manipulation's two configurations appear in Appendix E.1.

Additional execution configurations. Fixed-input evaluation also uses per-case limits of up to 48 code rounds and 480k tokens for Geometry and Affordance, 24 rounds and 384k tokens for non-VSI Spatial Reasoning, and 32 rounds and 960k tokens for VSI. Task time limits reach 7,200 seconds, or 10,800 seconds for VSI. Budgets vary by source and execution configuration.

Result selection. Fixed-input and Planning trial slots use the first valid outcome, independent of correctness. Manipulation uses the fixed adopted baseline and RGB-expanded manifests. The original Baseline CSV covers 182 binary assignments. Table 2 preserves every original model value and adds Opus. The Opus Manipulation score uses the same Baseline source

interfaces and is included in the main-table comparison. Appendix E.1 defines missing-record accounting, shared-current sources, and the common subset used in Table 3.

Coverage and scoring denominators. Table A7 retains the original cohort coverage. Opus metric-specific valid counts are in Table A12; no whole-domain valid count is inferred from overlapping summary metrics. HumanCLAW is excluded from the Opus Planning score pending completion and outcome reconciliation. Two RLBench exceptions and one RoboTwin exception count as unsuccessful assignments; Opus Manipulation remains 87/182 (47.8%). InFlux excludes four invalid references. Binary criteria retain all assigned tasks. Reported success requires the native evaluator, rather than an agent declaration. The inconsistent HumanCLAW record is not used in the Opus aggregate. Manipulation covers 182 binary tasks, including all five BEHAVIOR-1K assignments per model; AnymalC-Spin is excluded from this binary comparison. Continuous metrics retain finite valid predictions.

Table A7. Original seven-agent cohort. Valid submission coverage for fixed-input domains and recorded terminal outcomes for Planning. Denominators include repetitions; Manipulation uses the fixed allocation in Table A3.

Domain	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Geometry	1382/1402	1382/1402	819/886	1382/1402	1314/1402	1382/1402	1376/1402
Spatial Reasoning	1060/1100	1100/1100	586/660	1097/1100	1077/1100	1033/1100	982/1100
Affordance	350/350	350/350	208/210	350/350	335/350	349/350	349/350
Task Planning	157/157	157/157	157/157	157/157	157/157	157/157	157/157

Manipulation diagnostics. CALVIN reports average consecutive goals completed. Other auxiliary metrics average task-normalized native criteria or control evidence over fixed applicable subsets. They differ in their scoring items and evidence time windows; Appendix D.5 defines the aggregation, coverage, and missing-record treatment.

Main-table metric scope. Geometry rotation and translation pool the 30 Map-free and 24 GeoProbe pose cases in degrees and centimeters. Spatial Pass@1 covers all 220 cases; numerical AbsRel covers 27, and categorical accuracy separately covers 183. Affordance mask IoU and contact validity cover all three sources, with 70 cases. Planning uses terminal success across 157 cases for the original cohort and 144 non-HumanCLAW cases for Opus. Manipulation uses task-weighted native success across 182 binary assignments; Table A9 reports each source separately.

Dispersion and valid subsets. Pooled continuous statistics use $\bar{x} = K^{-1}\sum_{k=1}^{K} x_k$ and $v_{\text{pool}} = K^{-1}\sum_k (x_k - \bar{x})^2$ over valid case–run observations. This dispersion differs from the variance of run means in Equation 3. Main-table continuous metrics use pooled means; the task-level Geometry and Spatial analyses use equal-run means and SD, with common-valid-cohort checks in Appendices D.1 and D.2.

C Complete Model Comparisons

This section supports Section 4.2 at three resolutions: complementary native metrics, differences among source benchmarks, and the metric profiles shown in Figure 3. Task-level explanations follow in Appendix D.

C.1 Metrics and Source-Level Results

Table A8 gives complementary native metrics. Table A9 retains original-cohort source results, and Table A11 adds Opus, with denominators and repetition rules from Appendix B.4.

Table A8. Complementary native metrics across four domains. Repeated binary rates show mean ± run SD; single-run rates have no SD. Continuous values are pooled valid-observation means; Table A12 distinguishes case–run variance from variance of run means. †Opus Planning excludes HumanCLAW (73/144); the other agents retain 157 cases. N counts eligible cases. GP denotes GeoProbe; Manipulation source results appear in Table A9.

Metric	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	Opus
Geometry (370 cases)									
InFlux focal MAPE (%)↓	26	53.02	47.79	52.65	55.50	52.36	67.82	65.86	60.27
InFlux principal error (px)↓	26	3.82	3.82	3.82	3.82	3.88	3.82	3.82	3.82
Angle relative L2↓	12	0.760	0.915	1.32	1.41	0.962	1.30	2.04	1.166
Distance relative L2↓	10	0.294	0.418	0.785	0.560	0.419	0.411	0.331	0.455
Vector relative L2↓	8	0.425	0.591	0.625	0.454	0.647	1.02	1.03	0.516
Map-free rotation (°)↓	30	29.94	46.60	52.75	35.11	46.32	63.03	64.94	40.35
Map-free translation (cm)↓	30	188.21	265.85	259.75	243.06	280.99	430.82	374.96	240.27
Map-free direction (°)↓	30	26.28	48.91	53.02	38.92	52.42	95.15	78.78	43.56
GP focal MAPE (%)↓	36	6.92	9.57	31.71	14.67	26.59	33.39	38.56	16.48
GP principal error (px)↓	24	51.95	110.62	63.40	78.41	66.83	58.23	59.05	60.46
Depth MAE (m)↓	12	0.307	0.633	0.412	0.490	0.589	0.934	0.944	0.510
Depth RMSE (m)↓	12	0.338	0.753	0.517	0.598	0.699	1.05	1.06	0.599
Depth AbsRel↓	12	0.104	0.229	0.140	0.163	0.193	0.312	0.292	0.169
Displacement MAE (cm)↓	40	0.585	0.973	1.49	1.52	2.28	2.48	2.94	1.164
GP rotation (°)↓	24	1.17	2.54	2.83	2.21	3.10	7.55	5.26	3.05
GP translation (cm)↓	24	5.36	9.50	9.30	6.99	7.85	23.59	12.05	11.13
Scale MAPE (%)↓	44	1.90	2.84	3.76	3.77	3.93	16.07	16.45	3.91
Scale judgment Pass@1 (%)↑	12	100.0±0.0	100.0±0.0	100.0±0.0	100.0±0.0	100.0±0.0	100.0±0.0	93.3±7.0	100.0±0.0
2D trace RMSE↓	56	0.120	0.132	0.118	0.118	0.127	0.155	0.284	0.180
3D trace RMSE (m)↓	56	0.382	0.463	0.435	0.469	0.426	0.629	0.682	0.683
3D constraint success (%)↑	56	33.9	21.4	25.0	17.9	37.5	8.9	7.1	8.9
Spatial Reasoning (220 cases)									
Native Pass@1 (%)↑	220	69.4±1.0	60.7±2.5	62.0±4.6	68.9±1.1	65.4±1.3	50.6±1.8	48.0±0.9	58.6
Categorical Pass@1 (%)↑	183	82.2±0.6	71.3±3.0	72.7±5.2	81.2±1.3	76.7±1.9	60.0±2.2	56.8±0.7	69.4
Numerical AbsRel↓	27	0.213	0.273	0.186	0.205	0.161	0.251	0.288	0.222
Affordance (70 cases)									
BBox–GT box IoU↑	70	0.390	0.366	0.386	0.382	0.322	0.146	0.021	0.449
BBox–GT mask IoU↑	70	0.286	0.264	0.284	0.272	0.250	0.116	0.013	0.329
BBox success@0.5 (%)↑	70	46.6±1.3	43.7±1.6	46.2±0.8	45.7±1.7	33.1±11.8	5.4±3.7	0.3±0.6	54.3
Contact-point validity (%)↑	70	60.3±2.6	52.0±3.6	57.1±2.5	52.9±4.2	44.6±13.0	32.6±4.7	6.3±1.6	62.9
Task Planning (157 cases)									
Native Pass@1 (%)↑	157	80.3	44.6	63.7	35.7	48.4	17.8	19.1	50.7†

Table A9. Original seven-agent cohort. Source results. Spatial/Affordance: mean ± run SD (%); Planning: solved/assigned cases; Manipulation: success (%). N counts eligible cases.

Source	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Spatial Reasoning: categorical accuracy (%)								
3DSRBench	50	73.6 ± 2.6	71.2 ± 1.8	72.7 ± 3.1	74.0 ± 3.7	70.0 ± 2.4	69.2 ± 3.6	57.6 ± 2.6
BOP-ASK	20	96.0 ± 4.2	87.0 ± 4.5	88.3 ± 2.9	92.0 ± 2.7	97.0 ± 2.7	85.0 ± 3.5	84.0 ± 4.2
MindCube	50	90.8 ± 1.1	68.0 ± 7.3	79.3 ± 8.1	90.8 ± 2.7	89.2 ± 3.0	46.0 ± 2.8	50.4 ± 4.1
MMSI-Bench	20	77.0 ± 5.7	55.0 ± 11.7	66.7 ± 7.6	55.0 ± 8.7	56.0 ± 9.6	35.0 ± 9.4	29.0 ± 7.4
RoboSpatial	20	85.0 ± 3.5	81.0 ± 8.9	88.3 ± 2.9	88.0 ± 5.7	76.0 ± 4.2	90.0 ± 3.5	92.0 ± 5.7
VSI-Bench	23	72.2 ± 7.3	70.4 ± 5.7	36.2 ± 28.9	83.5 ± 8.4	65.2 ± 5.3	44.3 ± 6.4	39.1 ± 3.1
Affordance: strict box success at IoU ≥ 0.5 (%)								
ReasonAff-style	20	19.0 ± 5.5	6.0 ± 4.2	16.7 ± 7.6	13.0 ± 5.7	5.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0

Source-level comparisons, continued.

Source	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
RAGNet-3DOI	20	14.0 ± 4.2	15.0 ± 5.0	13.3 ± 2.9	7.0 ± 2.7	6.0 ± 2.2	4.0 ± 2.2	1.0 ± 2.2
UMD	30	86.7 ± 3.3	88.0 ± 4.5	87.8 ± 5.1	93.3 ± 2.4	70.0 ± 27.0	10.0 ± 7.8	0.0 ± 0.0
Task Planning: terminal success (single run)								
ALFRED	20	15/20	3/20	7/20	3/20	2/20	1/20	0/20
ALFWorld	20	14/20	4/20	6/20	2/20	4/20	3/20	4/20
DiscoveryWorld	10	10/10	4/10	7/10	3/10	5/10	1/10	1/10
HumanCLAW	13	3/13	2/13	1/13	0/13	3/13	0/13	0/13
ScienceWorld	73	64/73	38/73	59/73	29/73	45/73	11/73	14/73
VirtualHome	21	20/21	19/21	20/21	19/21	17/21	12/21	11/21
Total	157	126/157	70/157	100/157	56/157	76/157	28/157	30/157
Manipulation: baseline verified success (%)								
CALVIN	10	30.0	0.0	0.0	0.0	0.0	0.0	0.0
CLIPort	18	61.1	27.8	33.3	22.2	27.8	22.2	16.7
RLBench	10	60.0	20.0	10.0	0.0	20.0	0.0	10.0
RoboCasa	10	10.0	0.0	0.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	10.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	38.0	10.0	14.0	4.0	8.0	6.0	8.0
RoboWits	10	10.0	10.0	10.0	10.0	20.0	0.0	10.0
VIMA-Bench	10	40.0	30.0	30.0	10.0	20.0	30.0	30.0
VLABench	25	72.0	52.0	60.0	20.0	36.0	24.0	24.0
ManiSkill	9	55.6	0.0	–	0.0	11.1	0.0	0.0
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	182	41.8	19.8	24.7†	9.3	18.7	9.9	11.0

†Coverage, source configurations, and aggregation are specified in Appendix E.1.

Reading the metric families. GeoProbe focal MAPE combines full-intrinsics and focal-only queries; displacement MAE combines its four camera/environment conditions. Scale MAPE combines object dimensions and numerical reference-scale queries, while qualitative scale judgments retain Pass@1. TraceSpatial reports 2D and 3D errors separately because their coordinates and units differ. Each metric retains the target's physical interpretation.

C.2 Sensitivity to Source Composition

Table A10. Original seven-agent cohort. Task-composition analysis (success %). Spatial uses only the 183 categorical queries; Planning removes ScienceWorld and VirtualHome; Affordance removes UMD and uses strict box success. All columns retain the assigned-case denominator of the indicated subset.

Agent	Spatial without VSI	Planning without text/symbolic	Affordance without UMD
Astra	83.6	66.7	16.5
Sol	71.4	20.6	10.5
Fable	77.9	33.3	15.0
Qwen-Max	80.9	12.7	10.0
Gemini	78.4	22.2	5.5
Qwen-397B	62.2	7.9	2.0
Qwen-27B	59.4	7.9	0.5

Removing ScienceWorld and VirtualHome changes Astra's Planning success from 80.3% to 66.7%, and Qwen-Max's from 35.7% to 12.7%. The same table isolates categorical Spatial

Reasoning without VSI-Bench and strict affordance-box success without UMD. These subset comparisons measure sensitivity to the task mixture.

Equal source weighting. For the original seven-agent cohort, to test whether source allocation drives the lead, we average complete source scores with equal weight, $M_{\text{source}} = B^{-1} \sum_{b=1}^{B} M_b$, retaining every assigned case within each source. Across all six Spatial sources, Astra scores 71.9% versus Qwen-Max's 69.1%; across all three Affordance sources, contact validity is 54.2% versus Fable's 50.9%; across all six Planning sources, success is 75.2% versus Fable's 53.1%. These are the highest non-Astra macro scores within the original seven-agent cohort. The ranking therefore persists under both case weighting and equal source weighting. These analyses use the complete domain allocations, separate from the 46-case multi-round review diagnostic in Section 4.4.

C.3 Opus Repetitions, Dispersion, and Coverage

Opus was assigned five runs on the original 258 Geometry cases; tracing and all other domains have one run. Four invalid InFlux reference cases are excluded from eligible scoring. Continuous means and variances pool valid case–run observations. Population variance uses divisor K and squared metric units; it is not a standard error or a run-to-run SD. Table A12 retains the effective sample counts. For example, pooled pose rotation is 23.3°, while the equally weighted mean of five run means is 23.4°; differing valid counts explain the distinction.

Additional Opus results and scope. Planning excludes the 13 HumanCLAW assignments pending outcome verification. The remaining five sources yield 73 successes over 144 assignments (50.7%). The original agents retain their 157-case scores. Opus Manipulation uses the same Baseline protocol as the original main table; its 87/182 score counts two RLBench exceptions and one RoboTwin exception as unsuccessful assignments. The original paired-protocol analysis is retained separately. The supplemental RGB-expanded recordings in Appendix H illustrate individual episodes and are separate from this Baseline aggregate.

Table A11. Additional Opus source results. Planning excludes HumanCLAW. Manipulation retains all binary assignments, including execution errors. Spatial values cover all questions in each source, not only categorical questions.

Domain	Source	Assigned	Success (%)
Spatial	3DSRBench	50	60.0
Spatial	BOP-ASK	20	95.0
Spatial	MindCube	50	74.0
Spatial	MMSI-Bench	20	45.0
Spatial	RoboSpatial-Home	30	53.3
Spatial	VSI-Bench	50	36.0
Planning	ALFRED	20	15.0
Planning	ALFWorld	20	45.0
Planning	DiscoveryWorld	10	30.0
Planning	ScienceWorld	73	58.9
Planning	VirtualHome	21	71.4
Planning	Total without HumanCLAW	144	50.7
Manipulation	CALVIN	10	10.0
Manipulation	CLIPort	18	61.1
Manipulation	RLBench	10	30.0

Domain	Source	Assigned	Success (%)
Manipulation	RoboCasa	10	0.0
Manipulation	RoboCasa365	10	10.0
Manipulation	RoboWits	10	10.0
Manipulation	RoboTwin 2.0	50	46.0
Manipulation	VIMA-Bench	10	100.0
Manipulation	VLABench	25	76.0
Manipulation	ManiSkill	9	55.6
Manipulation	LIBERO-PRO	10	80.0
Manipulation	Robosuite	5	100.0
Manipulation	BEHAVIOR-1K	5	0.0
Manipulation	All assigned	182	47.8

Table A12. Opus continuous statistics. Variance is over valid case–run observations (population divisor). Coverage is valid/eligible; it includes repetitions.

Metric	Mean	Variance	Coverage
InFlux focal MAPE (%)	60.27	641	130/130
InFlux principal error (px)	3.82	27.7	130/130
Angle relative L2	1.166	3.59	60/60
Distance relative L2	0.455	0.787	50/50
Vector relative L2	0.516	0.267	40/40
Map-free rotation (°)	40.35	1.52e+03	142/150
Map-free translation (cm)	240.27	2.99e+04	142/150
Map-free direction (°)	43.56	1.67e+03	142/150
GP focal MAPE (%)	16.48	341	180/180
GP principal error (px)	60.46	2.35e+03	120/120
Depth MAE (m)	0.510	0.167	60/60
Depth RMSE (m)	0.599	0.179	60/60
Depth AbsRel	0.169	0.0167	60/60
Displacement MAE (cm)	1.164	2.2	200/200
GP rotation (°)	3.05	26.4	120/120
GP translation (cm)	11.13	378	120/120
Scale MAPE (%)	3.91	48.9	220/220
2D trace RMSE	0.180	0.0207	56/56
3D trace RMSE (m)	0.683	0.147	56/56
Numerical AbsRel	0.222	0.052	27/27
BBox–GT box IoU	0.449	0.151	70/70
BBox–GT mask IoU	0.329	0.0897	70/70
ReasonAff / BBox IoU	0.240	0.131	20/20
ReasonAff / Mask IoU	0.203	0.104	20/20
RAGNet-3DOI / BBox IoU	0.222	0.133	20/20
RAGNet-3DOI / Mask IoU	0.172	0.0784	20/20
UMD / BBox IoU	0.740	0.0271	30/30
UMD / Mask IoU	0.517	0.0252	30/30

Table A13. Opus Geometry variation across five run means. Variance is the sample variance of the five run means (divisor four), rather than pooled case–run variance.

Metric	Mean of runs	Run variance	Valid counts (R1–R5)
Pose rotation (°)	23.4	13.62	47/54/54/53/54
Pose translation (cm)	135.8	162.3	47/54/54/53/54
InFlux focal MAPE (%)	60.27	2.232	26/26/26/26/26
InFlux principal error (px)	3.82	0	26/26/26/26/26
Angle relative L2	1.166	0.01284	12/12/12/12/12
Distance relative L2	0.455	0.004787	10/10/10/10/10
Vector relative L2	0.516	0.001763	8/8/8/8/8
Map-free rotation (°)	40.95	65.58	23/30/30/29/30
Map-free translation (cm)	243.1	1283	23/30/30/29/30
Map-free direction (°)	44.29	82.1	23/30/30/29/30
GP focal MAPE (%)	16.48	37.63	36/36/36/36/36
GP principal error (px)	60.46	17.03	24/24/24/24/24
Depth MAE (m)	0.51	0.08685	12/12/12/12/12
Depth RMSE (m)	0.5988	0.09093	12/12/12/12/12
Depth AbsRel	0.1695	0.01156	12/12/12/12/12
Displacement MAE (cm)	1.164	0.2254	40/40/40/40/40
GP rotation (°)	3.048	0.9234	24/24/24/24/24
GP translation (cm)	11.13	2.178	24/24/24/24/24
Scale MAPE (%)	3.909	7.49	44/44/44/44/44

C.4 Interpreting the Capability Profiles

Figure 3 shows native metric values on shared axes within each panel; lower error is better for Geometry and numerical Spatial Reasoning, whereas higher overlap or success is better elsewhere. We avoid a single overall index because the metrics have different units and Opus Planning covers a smaller task subset. All displayed values are generated from the same eight-agent table as Table 2. Missing repetitions are not converted to zero uncertainty.

D Task-Level Analysis of Astra

The subsections follow Section 4.3 and Figures 4–8. These subtask diagnostics retain the original seven-agent cohort: the new summary does not supply Opus values for every subtask, and none are inferred from selected examples. Each first identifies the task requirement behind the main-text claim, then gives the supporting comparison and broader task coverage. Repetition and valid-coverage conventions are defined in Appendix B.4.

D.1 Geometry: Task Requirements and Source Conditions

Task-level comparisons. Table A14a supplies the absolute errors behind Figure 4a. Its relative reduction is $100(1 - e_{\text{Astra}}/e_{\text{peer}})$, using the lowest-error non-Astra peer for each target. Equal-run means retain task-specific units: MAPE for focal length, size, and height, and $100 \times \text{AbsRel}$ for depth. Astra leads the five GeoProbe targets; Sol leads on InFlux focal estimation.

Coverage sensitivity. Table A14b restricts each Astra–peer comparison to cases valid in every shared run. All six comparison directions remain unchanged. The ranking is thus stable on the common valid cohort.

Table A14. Geometric errors (%, lower is better), mean ± run SD. (a) Seven-agent comparison; GP denotes GeoProbe. (b) Common-valid cohorts with matched cases and runs.

(a) All valid cases

Task	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
GP focal-only	6.2±2.1	11.2±5.5	29.5±1.8	13.8±3.1	23.7±12.9	38.6±3.2	38.4±5.0
InFlux focal	53.0±2.1	47.8±2.0	52.6±2.2	55.5±1.0	52.4±3.4	67.8±1.8	65.9±2.1
GP depth	10.4±2.9	22.9±2.7	14.2±2.7	16.3±1.3	19.2±2.1	31.2±2.8	29.2±1.3
GP object size	6.3±1.0	8.5±0.7	7.6±0.2	9.6±0.8	8.3±0.3	11.4±0.6	9.4±0.3
GP ref. height	0.31±0.11	0.90±0.36	2.31±0.08	1.69±0.72	2.73±0.75	17.55±1.22	13.69±0.86
GP height ratio	0.32±0.15	0.95±0.41	2.23±0.17	2.17±0.22	2.67±0.28	20.35±0.87	17.08±2.70

(b) Matched common-valid cohorts

Task	Cases	Rounds	Peer	Astra	Peer
GP focal-only	12	5	Sol	6.21±2.10	11.16±5.47
InFlux focal	26	5	Sol	53.02±2.15	47.79±2.00
GP depth	9	3	Fable	10.16±3.49	14.22±2.88
GP object size	12	3	Fable	6.44±1.36	7.63±0.18
GP ref. height	12	5	Sol	0.31±0.11	0.90±0.36
GP height ratio	12	5	Sol	0.32±0.15	0.95±0.41

Camera motion and real photographs. The rendered displacement subsets share 12 object interventions while changing the camera condition. Table A15 compares Astra with Sol, the lowest-MAE non-Astra agent under each displacement condition. The fixed-camera advantage persists when the two stationary Blender cases are excluded. Under the native direction-and-1 cm criterion, both agents achieve 96.7% fixed-camera success: the continuous error distinguishes precision that this tolerance does not resolve.

The four real-photo pairs form a separate tabletop subset. Their displacement labels are supplied annotations, without independent camera or metric calibration. The Blender comparison uses different scenes and object motions.

Table A15. Displacement MAE (cm), mean ± run SD. Sol is the strongest non-Astra comparator under each listed condition. Moving-only excludes stationary cases.

Condition	Cases	Astra	Sol
Blender: fixed camera	12	0.240±0.109	0.467±0.267
Blender: fixed, moving-only	10	0.288±0.131	0.560±0.321
Blender: small view change	12	0.567±0.158	0.678±0.152
Blender: large view change	12	1.125±0.144	1.958±0.295
Real photographs	4	0.050±0.068	0.415±0.236

Coupled motion and metric scale reveal distinct estimation errors. Table A16 separates the GeoProbe pose modes: mixed rotation and translation produce 14.16 cm translation error and zero joint success, despite much lower errors on simpler motions. The best non-Astra translation estimate on the mixed subset is Qwen-Max at 16.77 cm; no evaluated agent achieves joint success on this subset. Translation-only cases have zero target rotation. Depth also separates ordering from metric scale: Astra's ordinal accuracy is 96.3%, whereas raw AbsRel is 0.104; oracle scale alignment reduces it to 0.038. Alignment uses reference depth.

Explicit scale references and regular rendered geometry support these estimates.

Table A16. GeoProbe camera-pose modes. Errors are pooled valid-observation means; joint success uses 2°/2 cm tolerances and shows mean ± run SD.

Motion	Cases	Rotation (°)	Translation (cm)	Joint success (%)
Stationary	4	0.00	0.00	100.0±0.0
Rotation only	6	0.35	1.00	96.7±7.5
Translation only	6	0.00	1.55	76.7±25.3
Rotation + translation	8	3.26	14.16	0.0±0.0

TraceSpatial-Bench: path matching and boundary constraints. Astra has lower 3D path RMSE than Gemini (0.382 versus 0.426 m), but fewer jointly valid paths (19/56 versus 21/56; Table A8). Among Astra's trajectories, 27 fail initiation, 24 fail arrival, and ten fail collision checks; these sets overlap. Of its 37 joint failures, 36 violate initiation or arrival, while only one satisfies both boundaries and fails collision avoidance alone. This supports the boundary-anchoring diagnosis in Figure 4c. The offline tolerances and sampled-position collision rule are defined in Appendix B.3.

D.2 Spatial Reasoning: Cross-Source Task Contrasts

Complete task coverage. Table A18 covers all 43 groups across six sources. Source categories are retained; MindCube uses task axes and question templates, prioritizing composed rotation and turn–translation before individual motion and relation queries.

Figure labels and comparisons. Figure 5a shows BOP-ASK left/right (L/R), MindCube relations (Rel.) and camera translation (Shift), 3DSRBench actor-relative side (Actor) and object orientation (Facing), and VSI hard directions (Hard). Panel b compares BOP camera depth (Near/Far), 3DSR camera/object distance (Cam./Obj.), and VSI relative distance (Rel.); each displayed peer leads or ties on a row. Source scenes are retained; the Obj. row has two cases.

Shared-round gaps. Panel c selects one task per source and its highest-mean non-Astra peer. Gaps and SD then use shared run indices (Table A17; protocol in Appendix B.3). On object-facing queries, Fable's three-run mean is 72.2% versus Astra's 50.0%; Astra averages 56.7% across all five runs.

Table A17. Source-specific Astra–peer comparisons for Figure 5c. Accuracy and gaps use shared available runs and are in percentages and percentage points. Run-index alignment follows Appendix B.3.

Task	Peer	Rounds	Astra	Peer	Gap ± SD
MMSI camera–region	Fable	3	100.0	77.8	+22.2±19.2
VSI medium direction	Qwen-Max	5	100.0	83.3	+16.7±11.8
BOP-ASK left/right	Gemini	5	95.0	96.7	−1.7±7.0
MindCube camera shift	Qwen-Max	5	78.5	89.2	−10.8±14.0
3DSR object facing	Fable	3	50.0	72.2	−22.2±9.6
RoboSpatial configuration	Q-397B	5	78.0	100.0	−22.0±4.5

Numerical error and acceptance are complementary. The numerical rows in Table A18 use the exact-match criterion from Appendix B.3. Table A19a instead measures continuous error on valid predictions. Fixing Astra's cohort across runs preserves low object-size AbsRel (0.091), while absolute-distance coverage is only 25/45 submissions (Table A19b). The table reports error together with valid-prediction coverage.

Table A18. All Spatial task groups, separated by benchmark. Entries are mean ± run SD (%); bold means mark the highest score in each row, including ties. Astra's column is shaded. N counts distinct cases; each assigned case remains in the success denominator.

Task	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
3DSRBench								50 cases
World height	4	60.0±13.7	50.0±0.0	**66.7**±14.4	55.0±11.2	25.0±0.0	50.0±0.0	50.0±0.0
Above / below	7	**77.1**±7.8	60.0±12.0	71.4±0.0	**77.1**±12.8	74.3±12.0	74.3±6.4	68.6±12.0
Camera depth	10	90.0±7.1	98.0±4.5	93.3±5.8	90.0±0.0	92.0±4.5	96.0±5.5	**100.0**±0.0
Next to	1	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	**60.0**±54.8	0.0±0.0
Object distance	2	70.0±27.4	80.0±27.4	83.3±28.9	**100.0**±0.0	**100.0**±0.0	50.0±0.0	80.0±27.4
Parallel / perpendicular	1	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	20.0±44.7	**100.0**±0.0	80.0±44.7
Same/different facing	4	55.0±11.2	50.0±0.0	50.0±0.0	50.0±0.0	50.0±0.0	**75.0**±17.7	35.0±13.7
Object-facing side	6	56.7±9.1	53.3±13.9	**72.2**±9.6	66.7±0.0	66.7±0.0	36.7±18.3	23.3±9.1
In front of actor	4	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	95.0±11.2
Actor left/right	7	**74.3**±6.4	71.4±14.3	61.9±8.2	71.4±10.1	65.7±7.8	45.7±12.0	5.7±7.8
Object viewpoint	4	**70.0**±11.2	**70.0**±11.2	50.0±0.0	60.0±13.7	**70.0**±11.2	**70.0**±11.2	65.0±28.5
BOP-ASK								20 cases
Closer to camera	5	96.0±8.9	96.0±8.9	86.7±11.5	**100.0**±0.0	96.0±8.9	80.0±14.1	84.0±8.9
Farther from camera	3	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	66.7±0.0	40.0±14.9
Left / right	12	95.0±4.6	80.0±4.6	86.1±4.8	86.7±4.6	**96.7**±4.6	91.7±0.0	95.0±4.6
MindCube								50 cases
Behind observer	7	97.1±6.4	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	88.6±12.0	80.0±16.3	65.7±16.3
Camera translation	13	78.5±10.0	64.6±8.8	56.4±16.0	**89.2**±6.9	84.6±5.4	20.0±6.9	35.4±12.9
Imagined observer	2	**90.0**±22.4	70.0±27.4	50.0±0.0	80.0±27.4	70.0±27.4	30.0±27.4	30.0±27.4
Object relations	12	**96.7**±4.6	63.3±12.6	91.7±8.3	85.0±3.7	95.0±7.5	31.7±7.0	35.0±3.7
Composed rotation	5	**100.0**±0.0	96.0±8.9	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	96.0±8.9
Turn then translate	11	**90.9**±6.4	43.6±7.6	75.8±10.5	**90.9**±6.4	87.3±10.4	49.1±13.8	58.2±10.4
MMSI								20 cases
Attribute (Appr.)	2	**90.0**±22.4	70.0±27.4	50.0±0.0	60.0±22.4	50.0±0.0	70.0±27.4	40.0±22.4
Attribute (Meas.)	2	**100.0**±0.0	50.0±0.0	66.7±57.7	40.0±22.4	50.0±0.0	0.0±0.0	40.0±22.4
MSR	2	40.0±41.8	30.0±27.4	**50.0**±0.0	40.0±22.4	30.0±27.4	30.0±44.7	40.0±22.4
Motion (Cam.)	1	80.0±44.7	80.0±44.7	**100.0**±0.0	**100.0**±0.0	60.0±54.8	0.0±0.0	0.0±0.0
Motion (Obj.)	1	**100.0**±0.0	80.0±44.7	**100.0**±0.0	80.0±44.7	80.0±44.7	80.0±44.7	20.0±44.7
Camera–camera relation	3	**100.0**±0.0	73.3±27.9	**100.0**±0.0	66.7±0.0	73.3±14.9	26.7±14.9	33.3±0.0
Camera–object relation	1	0.0±0.0	40.0±54.8	**66.7**±57.7	0.0±0.0	60.0±54.8	40.0±54.8	0.0±0.0
Camera–region relation	3	**100.0**±0.0	60.0±14.9	77.8±19.2	66.7±0.0	73.3±14.9	66.7±23.6	53.3±18.3
Object–object relation	1	**100.0**±0.0	80.0±44.7	**100.0**±0.0	60.0±54.8	20.0±44.7	80.0±44.7	60.0±54.8
Object–region relation	4	**50.0**±0.0	30.0±11.2	25.0±25.0	45.0±20.9	**50.0**±30.6	5.0±11.2	0.0±0.0
RoboSpatial								30 cases
Placement compatibility	10	92.0±4.5	92.0±8.4	**96.7**±5.8	90.0±0.0	64.0±8.9	80.0±7.1	90.0±7.1
Configuration	10	78.0±4.5	70.0±10.0	80.0±0.0	86.0±11.4	88.0±4.5	**100.0**±0.0	94.0±8.9
Free-space point set	10	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
VSI-Bench								50 cases
Appearance order	2	40.0±22.4	80.0±27.4	50.0±50.0	**100.0**±0.0	40.0±41.8	40.0±22.4	80.0±27.4
Absolute distance	9	2.2±5.0	2.2±5.0	3.7±6.4	0.0±0.0	**6.7**±6.1	0.0±0.0	2.2±5.0
Object count	5	32.0±11.0	20.0±14.1	**40.0**±20.0	36.0±16.7	36.0±16.7	28.0±11.0	12.0±11.0
Direction: easy	4	**100.0**±0.0	85.0±22.4	33.3±28.9	**100.0**±0.0	85.0±13.7	30.0±20.9	45.0±11.2
Direction: hard	4	45.0±11.2	30.0±27.4	25.0±43.3	**65.0**±28.5	60.0±22.4	30.0±32.6	5.0±11.2
Direction: medium	6	**100.0**±0.0	73.3±9.1	38.9±38.5	83.3±11.8	66.7±11.8	40.0±14.9	20.0±13.9
Relative distance	6	50.0±20.4	**76.7**±22.4	38.9±9.6	**76.7**±9.1	56.7±14.9	60.0±19.0	53.3±13.9
Object size	9	4.4±6.1	**22.2**±7.9	11.1±0.0	13.3±5.0	11.1±7.9	2.2±5.0	8.9±5.0
Room area	4	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
Route planning	1	**100.0**±0.0	**100.0**±0.0	33.3±57.7	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0

Table A19. VSI-Bench numerical estimation. (a) AbsRel, equal-run mean ± SD on each model's valid predictions; lower is better. (b) Astra's coverage and fixed-cohort sensitivity: common cases have valid predictions in every recorded round.

(a) Numerical error across agents

Task	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Object count	0.35±0.05	0.40±0.11	0.20±0.11	0.23±0.11	0.31±0.08	0.41±0.06	0.42±0.04
Absolute distance	0.28±0.11	0.42±0.05	0.28±0.07	0.26±0.02	0.12±0.03	0.29±0.09	0.31±0.08
Object size	0.09±0.01	0.09±0.02	0.10±0.01	0.12±0.02	0.05±0.01	0.13±0.02	0.19±0.02
Room area	0.24±0.03	0.19±0.03	0.21±0.04	0.26±0.07	0.27±0.04	0.22±0.05	0.26±0.10

(b) Astra coverage and common-cohort error

Task	Cases	Valid / assigned	Common cases	Common AbsRel
Object count	5	25/25	5	0.350±0.053
Absolute distance	9	25/45	2	0.279±0.083
Object size	9	42/45	8	0.091±0.017
Room area	4	20/20	4	0.240±0.029

Other task requirements. RoboSpatial separates placement compatibility (92.0%), configuration (78.0%), and free-space point sets (0.0%) for Astra. All agents fail the point-set criterion. Compatibility and configuration judgments succeed more often than point-set predictions.

D.3 Affordance: Contact and Region Extent

From valid contact to functional-region extent. Table A20 retains all fifteen action/hand groups, including zero-scoring conditions. UMD contact validity averages 96.7% for Astra and 94.4% for Fable. Cutting combines 100% valid contacts and 96% strict box success with mask IoU of 0.361; the box IoU is 0.694 (Figure 6b). This separates successful localization from the limitations of rectangular mask coverage. Containment adds a same-representation comparison: Astra's strict box success is 44%, versus Qwen-Max's 96% (Table A21).

Table A20. Astra Affordance metrics, mean ± run SD. IoU compares the native box to the GT mask; strict box success compares it to the GT box at IoU ≥ 0.5. Point and box success are percentages.

Action / hands	Cases	Mask IoU	Point (%)	Box (%)
RAGNet-3DOI				
free movement / two hands	2	0.046±0.063	20.0±27.4	20.0±27.4
free / one hand	1	0.384±0.351	60.0±54.8	60.0±54.8
free / two hands	5	0.249±0.070	60.0±14.1	36.0±16.7
pull / one hand	12	0.008±0.002	25.0±5.9	0.0±0.0
ReasonAff-style				
free movement / two hands	5	0.270±0.055	48.0±11.0	48.0±11.0
free / one hand	4	0.000±0.000	0.0±0.0	0.0±0.0
free / two hands	4	0.001±0.002	0.0±0.0	0.0±0.0
pull / one hand	4	0.098±0.033	60.0±13.7	0.0±0.0
push / one hand	3	0.318±0.221	46.7±29.8	46.7±29.8
UMD				
contain	5	0.472±0.067	100.0±0.0	44.0±16.7
cut	5	0.361±0.009	100.0±0.0	96.0±8.9
grasp	5	0.491±0.007	100.0±0.0	100.0±0.0
pound	5	0.637±0.009	100.0±0.0	100.0±0.0
scoop	5	0.504±0.054	80.0±0.0	80.0±0.0
support	5	0.638±0.017	100.0±0.0	100.0±0.0

Contact, extent, and shape give different comparisons. Astra leads mask IoU on UMD cutting, grasping, scooping, and support, while Sol leads containment and pounding. In the interaction-conditioned sources, one-handed pulling has zero strict box success for every model despite nonzero contact validity. Table A21 identifies the best peer separately for each reported criterion. ReasonAff and RAGNet retain their original scenes and interaction annotations.

Table A21. Metric-specific Affordance peers. Values are mean ± run SD, using the repetition counts in Appendix B.4. Peers have the highest non-Astra criterion means over available runs.

Task / criterion	Astra	Best peer	Model
UMD cutting: mask IoU	0.361±0.009	0.341±0.022	Fable
UMD cutting: strict box (%)	96.0±8.9	100.0±0.0	Fable
UMD containment: mask IoU	0.472±0.067	0.673±0.069	Sol
UMD containment: strict box (%)	44.0±16.7	96.0±8.9	Qwen-Max
ReasonAff pulling: point (%)	60.0±13.7	50.0±0.0	Fable
RAGNet pulling: point (%)	25.0±5.9	31.7±7.0	Qwen-Max
ReasonAff two-hand movement: point (%)	48.0±11.0	26.7±11.5	Fable
RAGNet two-hand movement: point (%)	20.0±27.4	50.0±0.0	Sol / Fable

D.4 Task Planning: Goal Families and Terminal States

Compound goals and source-specific requirements. Table A22 covers all 56 native families across the six sources. Each row compares Astra with the highest-success peer for that family, whereas Figure 7 keeps Fable fixed for source-level and household comparisons. Heating and dual placement test state change and multiple-object goal retention; ScienceWorld distinguishes experimental topics and instrument use. This preserves the task-specific advantages and reversals behind the aggregate Planning lead.

Table A22. Task Planning by native goal family, with one episode per assigned case. Entries are solved/assigned cases. One highest-success peer is shown per row; * denotes a tie with another peer. Source-level results for all seven models appear in Table A9.

Goal family	Astra	Best peer	Model
ALFRED			
Illumination	1/2	1/2	Fable*
Placement	4/4	2/4	Sol*
Container transport	2/2	1/2	Fable*
Cleaning	2/2	1/2	Sol*
Cooling	2/4	2/4	Fable
Heating	2/3	0/3	Sol*
Dual placement	2/3	0/3	Sol*
ALFWorld			
Illumination	3/3	2/3	Fable*
Placement	3/4	2/4	Sol*
Container transport	2/2	1/2	Gemini*
Cleaning	3/4	1/4	Sol*
Cooling	0/1	0/1	Sol*
Heating	1/1	0/1	Sol*
Dual placement	2/5	1/5	Sol*
DiscoveryWorld			
Archaeology dating	1/1	1/1	Sol*
Combinatorial chemistry	1/1	1/1	Sol*
It's not rocket science	1/1	1/1	Fable
Lost in translation	1/1	1/1	Fable
Plant nutrients	1/1	0/1	Sol*
Discovery feed	1/1	1/1	Sol*

Continued from previous page.

Goal family	Astra	Best peer	Model
Doors	1/1	1/1	Qwen-Max
Navigation in a house	1/1	1/1	Fable*
Pick and give	1/1	1/1	Sol*
Space sick	1/1	1/1	Fable
HumanCLAW			
Navigate to contact	3/5	3/5	Gemini
Sit on target	0/8	0/8	Sol*
ScienceWorld			
Find animal	3/3	3/3	Fable
Find living thing	2/2	2/2	Sol*
Find non living thing	3/3	3/3	Sol*
Find plant	3/3	3/3	Sol*
Grow fruit	2/3	2/3	Fable
Inclined plane determine angle	2/2	2/2	Fable
Inclined plane friction named surfaces	15/16	16/16	Fable
Inclined plane friction unnamed surfaces	1/1	1/1	Fable*
Lifespan longest lived	2/2	2/2	Sol*
Lifespan longest lived then shortest lived	1/1	1/1	Sol*
Lifespan shortest lived	1/1	1/1	Sol*
Measure melting point known substance	8/8	6/8	Gemini
Measure melting point unknown substance	1/1	1/1	Sol*
Mendelian genetics known plant	1/1	1/1	Sol*
Mendelian genetics unknown plant	3/5	1/5	Qwen-Max*
Test conductivity	6/7	6/7	Gemini
Test conductivity of unknown substances	6/8	6/8	Fable
Use thermometer	4/6	5/6	Fable*
VirtualHome			
Grab	1/1	1/1	Sol*
Greet	1/1	1/1	Sol*
Lie	1/1	1/1	Sol*
Navigation or observation	1/1	1/1	Sol*
Read	2/2	2/2	Fable*
Rinse	1/1	1/1	Sol*
Sit	3/3	3/3	Sol*
Switchoff	1/1	1/1	Sol*
Switchon	7/7	7/7	Sol*
Touch	1/1	1/1	Sol*
Type	1/1	0/1	Sol*
Wash	0/1	1/1	Sol*

Attempted actions and terminal goals. All eleven Astra household failures exhaust their assigned budget. ScienceWorld's nine failures have scored task-failure outcomes with no recorded parser-no-match, invalid-action, or code-exception events; negative score sentinels are not physical-progress measures. HumanCLAW identifies a different limit (Table A23): six Astra sitting episodes find the target, five approach within 20 cm and issue a sit action, yet none succeeds. Fable records no target-finding success on these episodes. The shared zero terminal rate conceals different progress before the required body–object condition.

D.5 Manipulation: Control and Protocol Dependence

Baseline performance and coverage. Table A24 supports Table 2 and Figure 8a with all thirteen sources and seven agents. The baseline aggregate combines ten-source baseline results with the three sources shared across conditions. Astra records 76 successes out of 182 assignments; Fable records 45. Appendix E.1 defines denominator and coverage.

Table A23. HumanCLAW sitting diagnostics for eight assigned cases per model. Counts pool independently recorded conditions. † denotes lower bounds because process fields are missing for two Gemini cases; dashes denote unavailable process metrics. Terminal success retains all eight assignments.

Model	Process coverage	Target found	Within 20 cm	Near + sit issued	Success
Astra	8/8	6	5	5	0
Sol	8/8	3	1	1	0
Fable	8/8	0	0	0	0
Qwen-Max	0/8	–	–	–	0
Gemini	6/8	3†	3†	3†	0
Q-397B	0/8	–	–	–	0
Q-27B	0/8	–	–	–	0

Consecutive goals and task-specific evidence. Table A25 reports auxiliary metrics under both evaluation conditions. CALVIN reports average consecutive goals completed. The other rows average task-normalized native criteria or control evidence, with definitions below.

Table A24. Complete Manipulation success by protocol (%). Baseline totals use 182 assignments; – marks missing records. The last three sources use the same records in both protocols. The common nine-source subset isolates sources with records under both protocols.

Benchmark	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Baseline								
CALVIN	10	30.0	0.0	0.0	0.0	0.0	0.0	0.0
CLIPort	18	61.1	27.8	33.3	22.2	27.8	22.2	16.7
RLBench	10	60.0	20.0	10.0	0.0	20.0	0.0	10.0
RoboCasa	10	10.0	0.0	0.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	10.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	38.0	10.0	14.0	4.0	8.0	6.0	8.0
RoboWits	10	10.0	10.0	10.0	10.0	20.0	0.0	10.0
VIMA-Bench	10	40.0	30.0	30.0	10.0	20.0	30.0	30.0
VLABench	25	72.0	52.0	60.0	20.0	36.0	24.0	24.0
ManiSkill	9	55.6	0.0	–	0.0	11.1	0.0	0.0
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
All assigned	182	41.8	19.8	24.7	9.3	18.7	9.9	11.0
Common nine sources	153	41.2	19.0	22.2	8.5	15.7	10.5	11.8
RGB-expanded								
CALVIN	10	0.0	10.0	40.0	10.0	0.0	0.0	0.0
CLIPort	18	55.6	33.3	50.0	27.8	38.9	22.2	16.7
RLBench	10	40.0	10.0	30.0	10.0	10.0	0.0	0.0
RoboCasa	10	10.0	0.0	10.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	20.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	42.0	12.0	34.0	14.0	6.0	4.0	6.0
RoboWits	10	10.0	10.0	20.0	20.0	10.0	0.0	10.0
VIMA-Bench	10	100.0	90.0	100.0	70.0	60.0	20.0	20.0
VLABench	25	64.0	48.0	68.0	32.0	52.0	24.0	28.0
ManiSkill	9	11.1	0.0	55.6	0.0	0.0	0.0	0.0
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
All assigned	182	39.6	23.6	44.5	19.2	22.0	8.8	9.9
Common nine sources	153	41.2	23.5	42.5	20.3	20.3	9.2	10.5

Table A25. Manipulation auxiliary metrics under both protocols. CALVIN reports average consecutive goals completed (0–5); other rows average task-normalized criteria or control evidence (%). Definitions and time windows appear in Table A26. – denotes unavailable records; the last three rows are shared.

Benchmark	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Baseline								
CALVIN goals (0–5)	10	3.10	1.40	0.90	0.80	1.80	0.00	0.10
RoboCasa	4	–	–	–	–	–	–	–
RoboCasa365	7	–	–	–	–	–	–	–
ManiSkill conditions	5	60.00	0.00	–	0.00	0.00	0.00	0.00
LIBERO-PRO	10	20.00	20.00	30.00	5.00	20.00	0.00	5.00
robosuite	5	60.00	60.00	80.00	40.00	80.00	30.00	16.67
BEHAVIOR-1K control	5	13.33	16.67	30.00	16.67	6.67	6.67	6.67
RGB-expanded								
CALVIN goals (0–5)	10	1.10	0.50	2.90	1.30	0.70	0.00	0.00
RoboCasa	4	25.00	33.33	83.33	50.00	50.00	16.67	25.00
RoboCasa365	7	33.33	23.81	57.14	28.57	47.62	28.57	23.81
ManiSkill conditions	5	0.00	0.00	60.00	0.00	0.00	0.00	0.00
LIBERO-PRO	10	20.00	20.00	30.00	5.00	20.00	0.00	5.00
robosuite	5	60.00	60.00	80.00	40.00	80.00	30.00	16.67
BEHAVIOR-1K control	5	13.33	16.67	30.00	16.67	6.67	6.67	6.67

Comparisons with matching records. The common nine-source subset excludes ManiSkill and the three shared-current sources, leaving 153 assignments per agent and protocol. Astra has 63 successes in both configurations, but task-specific gains and losses offset; Fable increases from 34 to 65. Appendix E.1 examines the corresponding execution evidence.

Metric definitions and coverage. The 183-case Manipulation inventory contains 182 binary tasks and one continuous-return AnymalC-Spin task. AnymalC-Spin is evaluated separately through continuous return. For each model and configuration, task success is $\mathrm{SR} = 100N^{-1}\sum_i s_i$, where $s_i = 1$ requires native acceptance. Runs without native acceptance receive zero success credit, including timeouts, execution or infrastructure failures, and missing records. Each task contributes one run to the assigned denominator.

CALVIN scores a five-goal sequence as successful only when all five goals are completed in order. Its average number of consecutive goals completed is $\bar{L} = 10^{-1}\sum_{i=1}^{10} L_i$, where $L_i \in \{0, \ldots, 5\}$ is the verified consecutive prefix. Unknown prefixes contribute zero: baseline Qwen-Max has one such record, while expanded Gemini's five available records contribute seven goals over the fixed ten-case denominator, giving 0.70.

For the other auxiliary metrics, case i has K_i scoring items and verified indicators z_{ij}. Its score is $u_i = K_i^{-1}\sum_{j=1}^{K_i} z_{ij}$; the reported mean is $100N^{-1}\sum_i u_i$. Failed or unknown items contribute zero. This weights tasks equally even when K_i differs. The diagnostics summarize recorded control events on fixed task subsets using the scoring items and evidence windows in Table A26.

Table A26. Six auxiliary metrics, averaging task-normalized evidence within each subset.

Source	N	Scoring items	Evidence window
RoboCasa	4	Approach, grasp and lift, stable release	Any verified event during execution
RoboCasa365	7	Same three transport milestones	Any verified event during execution
ManiSkill	5	Two or three native conditions per task	Final verification state
LIBERO-PRO	10	Two task-specific conditions	Final state and limited process records
robosuite	5	Two lifting, three stacking, or four wiping items	Native success and terminal reward-derived evidence
BEHAVIOR-1K	5	Six control-evidence items	Before the first episode-end signal

Transport control versus destination success. RoboCasa uses ArrangeBreadBasket, LoadDishwasher, PickPlaceToasterToCounter, and RecycleBottlesByType. RoboCasa365 uses BreadAndCheese, MakeBananaMilkshake, MicrowaveThawingFridge, MixCakeFrosting, PlaceMeatInMarinade, PlaceVegetablesEvenly, and RemoveCuttingBoardItems. Door, appliance, and navigation-only tasks are outside these subsets. Approach requires executed motion and confirmed arrival; grasp-and-lift requires both grasp and lift confirmation. Stable release requires prior grasp-and-lift of the same canonically named object, followed by confirmed release and stability. Each item scores at most once per case. These events need not cover every task object, and release does not establish the requested destination relation.

Table A27. ManiSkill's five-task diagnostic. Items are evaluated at final verification; grasp history is separate. The other four binary tasks remain in complete-task success.

Task	K_i	Scored conditions
PickCube	2	Object placed at its goal; placed and robot static. The suspended goal requires no release.
PlaceSphere	3	Sphere on its support; on support and released; both with the sphere static.
StackCube	3	Cube A on B; on B and released; both with cube A static.
PlugCharger	2	Goal distance ≤ 0.005 m; distance satisfied and angular error ≤ 0.2 rad.
PegInsertionSide	2	Peg-head $x \geq -0.015$ m in the hole frame; official insertion success.

LiftPegUpright, PullCube, PushCube, and RollBall contribute to the nine-task success rate but not this diagnostic. Dashes mark unavailable Fable baseline records in both sets.

Task-specific conditions in LIBERO-PRO and robosuite. LIBERO-PRO uses fixed init-0 instances: spatial 1/7, object 2/5, goal 1/7, libero_10 1/9, spatial_swap 0, and object_swap 4. Ordinary placement scores recorded grasp and the official target relation; stove opening scores operation of the correct control and an open stove. The two longer tasks score a cup inside a microwave followed by the door being closed, or one food item followed by both items in a basket. Missing grasp/control records score zero; for the longer tasks, necessary preceding relations are confirmed only on official success.

robosuite uses cube lifting with seeds 99/50, cube stacking and restacking with seed 14, and spill wiping with seed 48. Lifting scores recorded grasp and native success; grasp history is missing. Stacking uses the fixed local reward q: $q > 0.125$ supports a grasp/lift/stack condition, $q \geq 0.5$ supports the top cube being lifted, and native success supplies the third item. A restacking zero can mask intermediate events under the two-cube airborne rule; these items remain unknown and receive zero credit. Wiping checks 25%, 50%, 75%, and complete removal. Current records verify the first three only on native complete success, rather than interpreting total reward as a wiped fraction. Consequently, Fable's 100% complete-task success and 80% auxiliary score are compatible: missing intermediate evidence lowers the latter.

Table A28. BEHAVIOR-1K control evidence. Each item records one type of control event and scores once per case; terminal BDDL goal success is evaluated separately.

Item	Positive evidence
Localization	Finite coordinates or pose from public RGB-D localization or grasp sampling.
Candidate	A nonempty pregrasp or grasp candidate set.
Arrival	Measured arrival from approach or recovery feedback for the same arm.
Holding	An explicit in-hand check or holding confirmation inside checked placement.
Held motion	Associated measured motion of at least 5 cm while holding, or movement with a passed holding check inside checked placement.
Release	Holding, active gripper opening, then no longer holding; or confirmed release inside checked placement.

The five tasks are outfit_a_basic_toolbox, can_meat, tidying_bedroom, putting_shoes_on_rack, and chop_an_onion. Records are linked by arm, name, and temporal order; physical-instance IDs are not consistently recorded. Object identity and inside/ontop relations are checked separately by the terminal goal evaluator. Held motion uses a 5 cm threshold. Control-event records cover four tasks for each of four models and the toolbox task for each of the other three. All retain the five-task denominator. Scoring uses the evaluated episodes up to their first end signal and requires direct evidence for each item. Fable's nine verified items yield $100 \times 9/(5 \times 6) = 30\%$, versus Astra's four items and 13.33%. RoboWits reports task success.

E Execution and Multi-Round Review Comparisons

E.1 Manipulation under the RGB-Expanded Protocol

Protocol and denominators. Baseline uses Table 2's source-specific observations and native tools. RGB-expanded adds current RGB at each decision. Both conditions use matched inference budgets, interaction allowances, and native execution-step limits. The comparison covers ten suites and 162 tasks: CALVIN, CLIPort, RLBench, RoboCasa, RoboCasa365, RoboTwin 2.0, RoboWits, VIMA, VLABench, and ManiSkill. Table 3b displays three suites; Table A24 gives all results. Common excludes ManiSkill because Fable's baseline records are missing, retaining nine suites and 153 tasks. LIBERO-PRO, robosuite, and BEHAVIOR-1K reuse 20 current records in both CSVs, bringing the domain total to 182. Missing baseline Fable ManiSkill and RoboCasa/365 stage records appear as dashes; the domain success aggregate credits the missing ManiSkill records with zero successes.

Observation and interaction settings. Baseline and RGB-expanded share the per-source inference and interaction budgets in Table A29, together with the same native execution-step limits. Baseline uses each source's native observations and tools; RGB-expanded provides the current scene's RGB views at every model decision. VIMA also supplies task reference images. Text interaction history and public state/perception tools remain available. Both conditions share an agent loop in which one code response can execute several actions.

Request and token accounting. The audited 7,378 expanded Astra/Sol/Fable requests across ten ordinary groups use temperature zero, with neither explicit reasoning effort nor a separate thinking budget. Of these, 7,372 request 3,600 response tokens; six use smaller limits as the case budget runs low. The configurations use a 3,600-token response ceiling and a 12-minute request timeout. Reported reasoning usage is recorded separately and counted once in the case total, avoiding duplication when included in completion usage; absent usage details remain unknown. Verifier allowance is one more than the model-round limit.

Tool calls, action calls, and physical steps are distinct counters. Trial time includes model waiting, code execution, and feedback within the loop; Code and Native limit one code execution and one native call, respectively. Request timeouts do not add extra time beyond Trial. Initialization, outer watchdogs, and retry backoff have separate clocks, so Trial is not identical to total wall-clock duration.

VIMA's reference-image count varies by task. The remaining seven ManiSkill inventory cases, including AnymalC-Spin, use one base view. RoboCasa/365 views are rendered scene observations. Request images and process videos are separate records: continuous video is not enabled uniformly for VLABench, LIBERO-PRO, or robosuite, while the later BEHAVIOR revision queue records at 15 fps. Video frame rate does not define model observation frequency.

Table A29. Shared execution ceilings for Baseline and RGB-expanded. Times are minutes; token budgets are thousands. A round is a model decision, Tools counts interface calls, and Checks counts verifier calls. The limits apply to all seven agents within each source.

Source	Rounds	Tools	Checks	Tokens	Trial	Code	Native
CALVIN	40	7,000	41	1,280	120	10	10
CLIPort	24	256	25	960	60	10	10
RLBench	16	256	17	640	60	10	10
RoboCasa	24	256	25	1,120	60	6	10
RoboCasa365	24	256	25	1,120	60	8	10
RoboTwin 2.0	16	256	17	640	60	10	20
RoboWits	24	256	25	640	60	30	20
VIMA-Bench	16	256	17	640	60	10	10
VLABench	16	256	17	640	60	10	10
ManiSkill	32	1,024	33	640	60	10	10
LIBERO-PRO	24	256	25	640	180	60	60
robosuite	24	256	25	640	180	60	60
BEHAVIOR-1K	24	256	25	640	180	60	60

Table A30. Expanded camera presets, width × height in pixels. The preset inventory covers all 183 assigned cases, including AnymalC-Spin. Multiple views enter as separate images.

Source	Views	Resolution
CALVIN	Static; gripper	200×200; 84×84
CLIPort	Front, left, right; no Oracle camera	640×480
RLBench	Front, wrist, overhead	128×128
RoboCasa/365	Left/right agent view, eye-in-hand	512×512
RoboTwin 2.0	Head, left, right	320×240
RoboWits	Ego, left/right wrist	848×480
VIMA-Bench	Front/top scene views and task reference images	256×128
VLABench	Right, left, forward, Franka wrist	480×480
ManiSkill	Base; additional hand view for PegInsertionSide, PlugCharger, StackCube	128×128
LIBERO-PRO	Agent view, wrist	800×512
robosuite	Agent view, wrist	512×512
BEHAVIOR-1K	Ego, left/right wrist	512×512

CALVIN: interaction versus feedback control. The execution analysis reports Astra's average consecutive goals decreasing from 3.10 to 1.10, Sol's from 1.40 to 0.50, and Fable's increasing from 0.90 to 2.90. Astra's median action-interface calls fall from 1,795 to 480.5, and Sol's from 1,247.5 to 734. Their median cumulative prompt tokens rise from approximately 321k and 363k to 1,206k and 1,217k. These totals accumulate prompt tokens over successive requests. Under RGB-expanded, five Astra and eight Sol episodes exhaust the shared per-case token allowance. Sol nevertheless increases full-sequence success from zero to one, illustrating why mean progress and completion need separate reporting.

Figure A2a connects these aggregate changes to controller construction and final goal completion in sequence 0557.

Budget utilization and interface effects. Three of Fable's four RGB-expanded CALVIN successes finish at rounds 22, 24, and 25, within the shared execution allowance. These episodes use later rounds to continue control after establishing feedback. Figure A2b examines controller construction in PickCube. Tool semantics also matter: the RGB tool compares colors on a 0–255 scale, whereas some Astra/Sol calls use 0–1 values and tight tolerances, producing empty detections. Fable obtains the red region with [255,0,0] and tolerance 80. Correct units determine whether the controller receives a usable target.

Input and execution audit. The audit covers 7,378 Astra/Sol/Fable requests from the ten ordinary benchmark groups: all contain RGB, archived Astra/Sol responses have no length

Table A31. Native execution-step limits shared by both conditions. Values show environment defaults → effective evaluation limits; control frequency is unchanged.

Source	Base → effective	Counter and termination semantics
CLIPort	5 → 10	Pick-and-place actions
RoboCasa/365	1000 → 2000	Control steps; field does not truncate episodes in the current wrapper
RoboTwin 2.0	400 → 800	Native action calls, not all simulator steps
RoboWits	800 → 1600	Control steps for the checked case
ManiSkill	50 → 100	Control steps; PickCube termination/boundary checks verify the effective limit
LIBERO-PRO	4000 → 8000	Wrapper/control counter; field does not truncate episodes
robosuite	1500 → 3000	Wrapper/control counter; field does not truncate episodes
BEHAVIOR-1K	500 → 1000	Environment control steps; later collision revision retains 1000
CALVIN, RLBench, VIMA	No finite limit	Checked instances remain bounded by rounds, calls, tokens, and time
VLABench	Unlimited	Simulation-time limit remains unlimited

termination, and Fable has 396. RLBench still contains four Astra, two Sol, and four Fable code-execution timeouts. Astra/Sol failures therefore occur even with delivered images and complete responses; code timeouts and interface semantics provide additional execution-level explanations. VIMA improves for all three agents, and Astra improves on RoboTwin while remaining unchanged over the common 153 tasks. The gains vary with the task and how each model uses the additional visual observations.

E.2 Fixed-Input Multi-Round Review

Multi-round review lets agents revisit their initial answers using the same task observations and scoring criteria. Self-review adds reconsideration, Crop review adds image crops, and Tool-assisted review adds specialist visual and geometric measurements. The three conditions permit up to two, two, and six additional responses, respectively, across Geometry, Spatial Reasoning, and Affordance. Table 3a reports results; sampling and tools are specified below.

Shared cases. Astra and Qwen-27B share 46 cases: 19 Geometry, 18 Spatial Reasoning, and nine Affordance. A 72-case pilot covers 24 benchmark/task strata selected by task category and SHA-256 ordering with seed 20260923, independently of scores and answers. It excludes the 60 exact cases from an earlier performance-selected study. One case per stratum forms a 24-case development set; withholding two more cases that share development scenes leaves the 46 evaluation cases. Some scenes recur, and some cases were inspected during development.

Table A32. Review conditions for the same cases and native scoring interfaces. Budgets are additional to native execution; response limits count output tokens.

Condition	Evidence available during review	Additional response budget
Initial	Original task observations	None
Self-review	Original observations and initial answer	Two responses, up to 1,800 tokens each
Crop review	Self-review evidence plus image crops	Two responses, up to 1,800 tokens each
Tool-assisted	Original RGB, crops, and specialist measurements	Six responses, up to 4,096 tokens each; up to 16 specialist requests

Evidence construction. Tool-assisted review adapts GCA's measurement and reference-frame formalization (Chen et al., 2025b) and SpatialClaw's inspectable execution (Cho et al., 2026). It formulates the measurement before consulting the candidate answer, then uses a persistent Python session. SAM 3 (Carion et al., 2025) supplies regions; SuperPoint (DeTone et al., 2018) and LightGlue (Lindenberger et al., 2023) supply correspondences; GeoCalib (Veicht et al., 2024) estimates calibration. Numerical routines compute relations and estimates. Finite-value

and coordinate checks validate the computed outputs. Remaining semantic errors include selecting a whole-object mask for a functional part and ambiguous translation scale.

Observation and execution controls. Specialist tools receive only previously exposed RGB, without predicted depth, learned 3D reconstruction, private calibration, scoring annotations, or extra video frames. Public depth or masks originally supplied by a task remain available to the base model. Review reuses native candidates where available; one Astra Spatial case starts from its public observation because the candidate is missing. Temperature is zero, and total-token and wall-time accounting includes native execution. Incomplete runs may be retried; the first completed result is retained regardless of correctness, yielding one outcome per condition and case.

Outcome interpretation. Geometry counts improved, tied, and worsened results across 24 metric entries from 19 cases; one Qwen-27B crop entry is unavailable. Spatial accuracy retains 18 assigned cases, with missing answers counted as incorrect. Affordance reports box–box IoU and success at IoU≥ 0.5 over nine cases.

E.3 Native Capability and Complete Task Execution

The main study asks how far a VLM's own perception, reasoning, and action selection support embodied generalism through source-native interfaces. Its central difficulty is completing the task with sufficient precision before the execution budget runs out. Related robot-agent studies help distinguish contextual guidance and supplied control competence from the decisions the model must make itself.

Context and control competence. Robotic in-context learning connects demonstrations and interaction to execution through context-conditioned policies, geometric transfer, world models, or skill- and agent-based control (Huang et al., 2026a). Keeping model weights fixed therefore leaves the source of execution competence to be specified. Learned skill selection (Ahn et al., 2022), programmatic composition of perception and control APIs (Liang et al., 2022), and planning through spatial value maps (Huang et al., 2023a) assign different responsibilities to the model and its executor. GPT-Policy (Cheng et al., 2026) uses demonstrations, action references, and feedback to condition a fixed VLM, while a constrained controller checks and executes its requests. Our baseline retains source-native helpers but leaves task interpretation and action selection to the model; specialist measurements enter only the supplementary review study. This identifies which parts of execution test the model's native capability.

Task composition. Composite tasks require several operations while preserving achieved conditions. Our RoboCasa sample contains seven atomic and three composite tasks; RoboCasa365 contains three atomic and seven composite tasks (Appendix A). PyRUA-Lean (Si et al., 2026) evaluates 18 atomic and 32 composite RoboCasa365 task types at five seeds each, reporting 86.7% and 39.4% success. Task composition changes the completion challenge, and its Target50 tasks differ from our pretrain-designated instances.

External execution support. Learned VLA policies supply sensorimotor competence in addition to model-directed reasoning. Removing PyRUA-Lean's frozen RLDX-1 policy lowers atomic and composite success to 65.6% and 20.0%, respectively (Si et al., 2026). The remaining successes retain programmatic composition of other primitives. Our RoboCasa protocols allow 24 model rounds and 60 minutes of trial time; PyRUA-Lean allows 40 calls for atomic tasks, 100 for composite tasks, and a two-hour episode watchdog. Both support multiple actions per

Python response. These results depend on task instances, controller support, and execution allowances as well as the underlying VLM.

Extended interaction without completion. GPT-Policy (Cheng et al., 2026) reports 0/3 successful towel pickups without demonstrations, averaging 96.3 decisions and 24.6 minutes. Human video raises success to 2/3 while reducing these averages to 76.7 decisions and 18.9 minutes; averages include failed trials. Our trajectories expose a related interaction burden: five Astra and eight Sol RGB-expanded CALVIN episodes exhaust their token allowance, while three of Fable's four successes finish at rounds 22, 24, and 25 (Appendix E.1). In PickCube, Astra and Sol use 25 and 32 responses without success, whereas Fable succeeds through calibrated feedback control (Figure A2). Our round and token counts establish budget use; they do not substitute for measured elapsed time. Extended interaction alone does not establish successful task completion.

Precision and terminal goal satisfaction. More interaction also need not resolve the geometric conditions that determine success. In GPT-Policy's plug task, robot video yields 0/3 successes, while adding aligned action references yields 2/3; success requires the plug to remain fully seated after release (Cheng et al., 2026). In our arena, initiation or arrival errors occur in 36 of Astra's 37 failed traces despite its lowest trajectory RMSE (Figure 4). Five HumanCLAW episodes reach the seat and issue the sitting action without establishing the required sitting state (Appendix D.4). These findings make the limit concrete: native VLM competence can produce plausible local decisions and repeated actions while still falling short of precise, complete execution within a finite budget. Embodied generalism requires efficient control, accurate endpoints, and preservation of every required goal condition through termination.

F Case Studies

From spatial judgments to usable contacts. Figure A1 illustrates Section 4.3 with six cases from the first evaluation run.

① *Spatial references.* In these runs, Astra correctly judges placement compatibility but misidentifies object-facing direction and camera translation. ② *Contact versus coverage.* Astra identifies usable tool and furniture regions. For pulling, three models place valid contacts on the handle, but their boxes cover only a fraction of the reference door region.

Controller construction across observation settings. Figure A2 links controller construction to terminal outcomes in CALVIN sequence 0557 and ManiSkill PickCube; execution settings are in Appendix E.1.

CALVIN: Astra's feedback controller emerges later. Baseline Astra builds control from tool-center-point error at round 5; RGB-expanded Astra builds it at round 12. Fable constructs a state-reading movement loop at round 2. Earlier controller construction coincides with longer completed goal sequences in these cases.

PickCube: fewer steps, better organized control. Fable calibrates joint response before constructing reusable numerical feedback control, then succeeds in 36 native steps. Astra and Sol use 100 and 56 steps across 25 and 32 responses without success; their trajectories emphasize direct joint increments and cross-round queries.

Motion and task completion. Figure A3 contrasts visible motion with verified task completion.

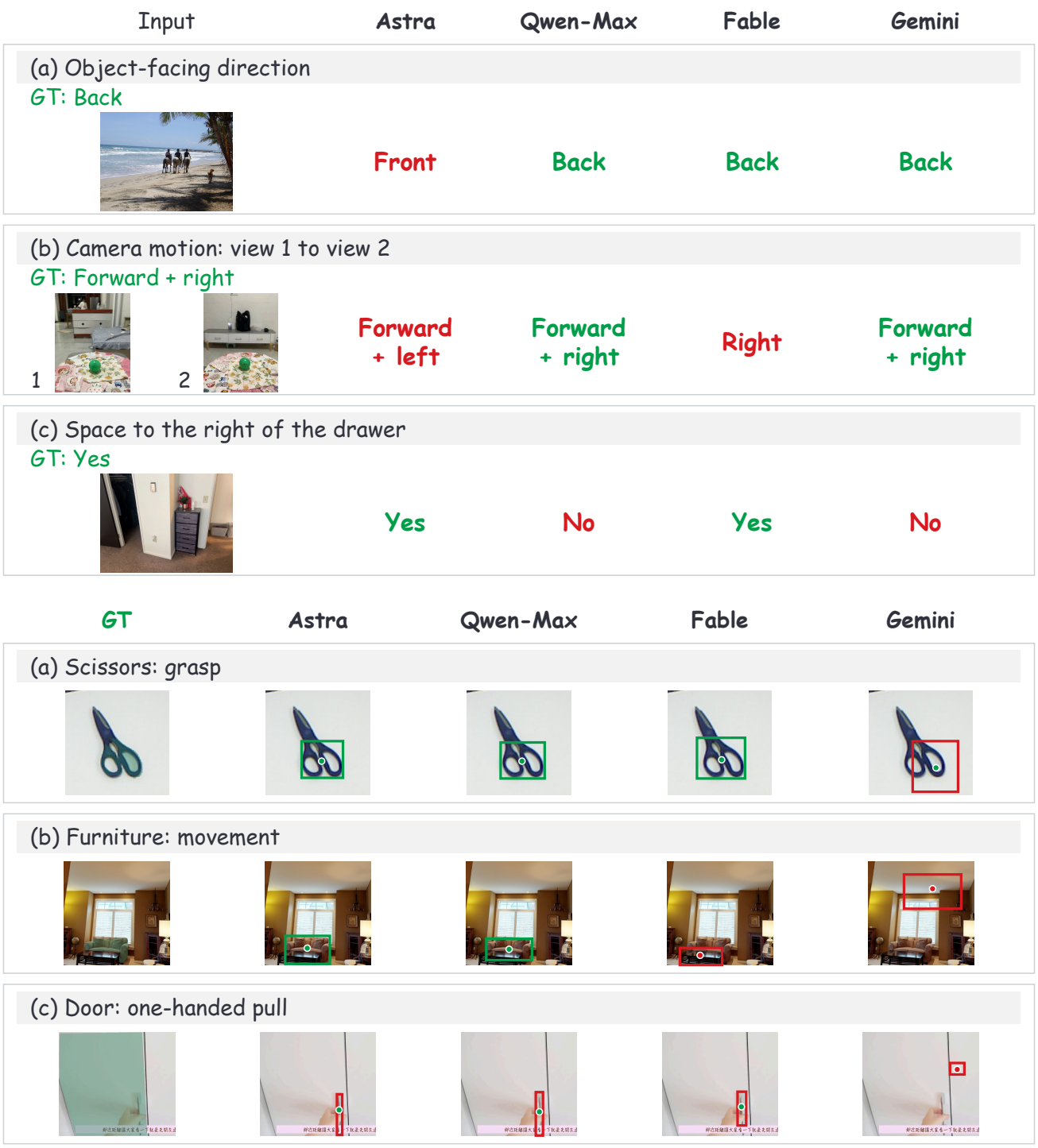


Figure A1. Spatial judgments from the first evaluation run (top: 3DSRBench, MindCube, RoboSpatial) and affordance localization (bottom: UMD, ReasonAff, RAGNet). Top questions concern the dog's facing direction, camera motion, and whether the plant fits beside the drawer; green answers match GT. Below, green points lie inside the GT mask and green boxes have IoU≥ 0.5; red denotes failure.

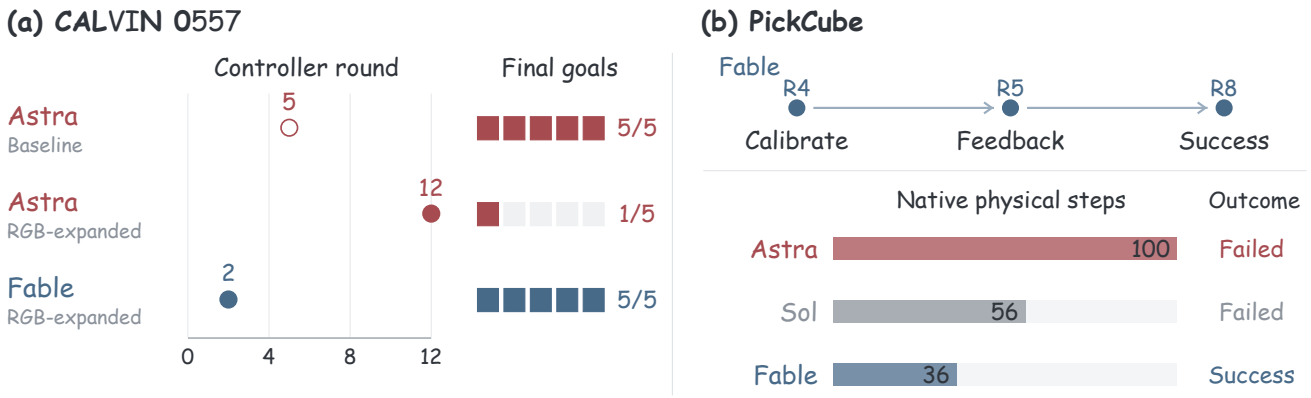


Figure A2. Controller construction and task outcomes. (a) CALVIN dots mark controller construction rounds; filled tiles count final consecutive goals. (b) PickCube shows Fable's ordered calibration, feedback-control, and success events, plus all three agents' physical-step totals and outcomes under RGB-expanded execution. R denotes model rounds; physical steps count environment updates.

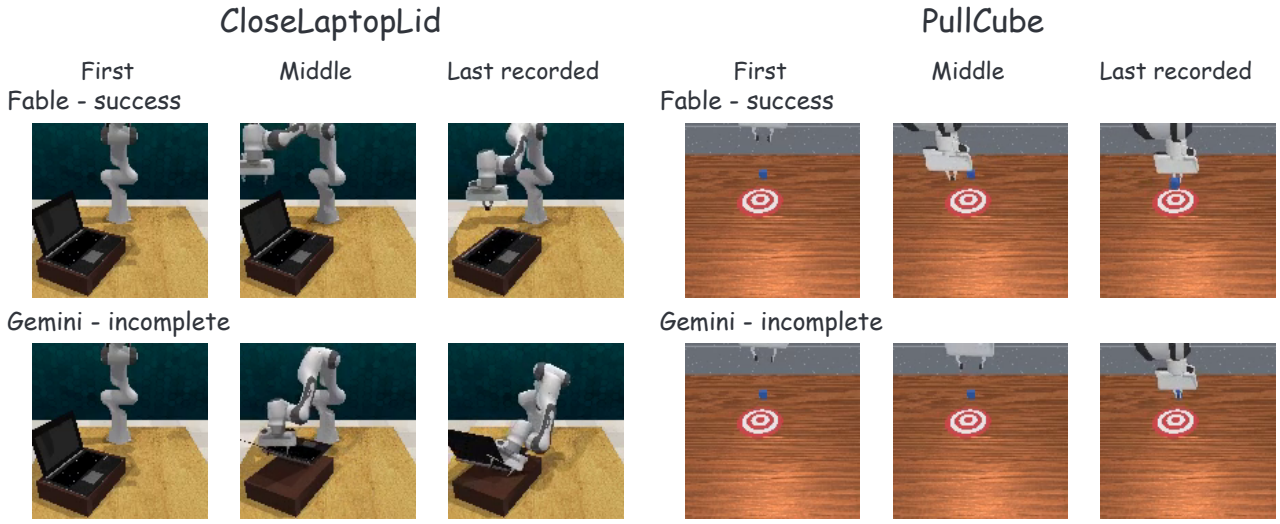


Figure A3. Archived execution contrasts under matched task settings and budgets. CloseLaptopLid uses RLBench's front camera; PullCube uses ManiSkill's base camera. Native records determine success. These earlier episodes are separate from the protocol comparison.

G Evaluation Scales and Public Resources

Table A33 expands Table 1 with instance counts and released artifacts. Task categories, training demonstrations, and source datasets are excluded from the instance count. Our release provides a fixed 1,000-case index across five capability domains, together with observations, episode initializations, and evaluation code.

OpenEQA adds navigation to episodic QA in its active track; Memory denotes cross-session persistence. VLABench's non-interactive track scores action sequences from fixed multi-view images against reference graphs. Hard Vision, Easy Vision varies reasoning effort and tool use; its 55 source benchmarks do not specify an instance count.

Table A33. Reported evaluation populations and released resources. Counts follow the cited papers; artifact availability is checked on 3–4 October 2026. Task-type and source counts describe coverage rather than tested instances. Links lead to author-maintained resources.

Study	Reported evaluation population	Released evaluation resources
OpenEQA (2024)	1,636 QA entries; the 557 HM3D questions also support active navigation. GPT-4V's active evaluation uses a 184-question subset.	Baseline and scoring code; QA manifest and episode-history download instructions.
BEAR (2025)	4,469 multimodal QA entries; robot online validation is additional.	Agent code; benchmark data and scoring scripts.
EmbodiedBench (2025a)	1,128 testing instances in four environments.	Code and configurations; benchmark datasets.
EmbodiedEval (2025)	328 tasks in 125 scenes.	Code; 328-entry test set and scene assets.
Embodied Arena (2025)	22 source benchmarks; a combined evaluated-instance count is not reported.	Platform and configurations; source datasets are downloaded separately.
CaP-X (2026)	Core: seven task types, each with 100 trials per tier. Extensions include 30 LIBERO-PRO task types and two BEHAVIOR tasks with 25 trials each.	Code, environments, and evaluation scripts. The repository lists 39 task types; the paper describes 187 platform tasks.
VLABench (2024)	100 task categories: 60 primitive and 40 composite; episode counts depend on the experiment.	Code and evaluation tracks; fixed-image VLM evaluation data.
WorldLines (2026b)	310 memory QA entries and 21 planning cases; the 62-entry ablation uses a QA subset.	The paper supplies prompts and examples; an author code/data release was not located.
Hard/Easy Vision (2026)	55 benchmarks, 34 capabilities, nine areas; total and per-benchmark evaluated-instance counts are not reported.	Results and qualitative examples; the repository contains a README and overview image, without evaluation scripts or a test manifest.
Embodied Agent Arena	**1,000 cases** from 32 established sources and GeoProbe, including 168 GeoProbe cases.	Harness and evaluation code; case index, observations, and episode initializations.

H Additional Cases Across the Five Capabilities

The examples in Figures A4–A8 connect an explicit task instruction to the model's output or recorded execution. They cover geometric estimation, spatial judgments, functional localization, and complete task outcomes. The accompanying online case atlas contains 129 selected examples across 43 benchmark entries and subsets, with three examples per entry. Selection emphasizes task variety, differing answers, and both successful and unsuccessful runs; these examples do not estimate aggregate accuracy.

Models and execution settings. The additional material includes Claude Opus 5.5 (recorded identifier `claude-opus-5-5`), now also represented in the updated quantitative snapshot. Selected cases illustrate individual records and do not replace the assigned-case denominators in Table 2. Geometry, Spatial Reasoning, and Affordance panels show each model's first recorded answer; the spatial panels use the first run without YOLOE, excluding later perception-assisted repetitions. In the execution panels, Opus uses the recorded source interfaces: ALFWorld visual primitives, DiscoveryWorld text/JSON operations, HumanCLAW egocentric RGB actions, and the RGB-expanded Manipulation interfaces. These supplemental Manipulation episodes are separate from the Baseline aggregate in Table 2. These interfaces may expose source-specific state or helper operations; the examples therefore do not establish a controlled model ranking or an improvement attributable solely to the model.

Reading the evidence. For static tasks, the reference and individual predictions are shown together. Affordance contact denotes membership in the reference mask; box IoU compares the predicted and reference rectangles. Neither is a generated segmentation mask. For execution tasks, the outcome is the terminal run report, not a judgment made from an attractive video frame. Planning panels contain ordered saved observations; the DiscoveryWorld images visualize the environment rather than the agent's text/JSON input. Manipulation panels sample one recorded camera stream; their timestamps index encoded video, not wall-clock execution. Elapsed runtime is reported separately. A failed terminal state and exhaustion of a model-round limit are distinct stopping conditions.

The cases expose a common gap between partial competence and task completion. Correctly estimating one displacement does not ensure a correct camera-frame transformation; locating a contact does not ensure adequate region coverage; and visible motion does not establish completion of every task predicate. The execution records make this distinction concrete: a HumanCLAW episode ends unsuccessfully before its time ceiling, whereas the CALVIN and LIBERO-PRO failures shown below reach their respective model-round limits.

(a) GeoProbe | Real Object Motion

Track the green-handled glass jug relative to the table.

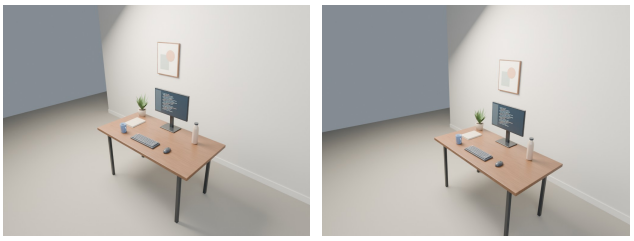
View 1 View 2

Model	Direction / cm
GPT-6 Astra	right / 2
GPT-6 Sol	right / 2
Fable 5.1	right / 4
Gemini 3.8 Flash	none / 0
Qwen 3.8 Max	right / 3
Qwen 3.5 397B	none / 0
Qwen 3.5 27B	left / 7
Opus 5.5	right / 2.5

Reference: right, 2 cm

(b) GeoProbe | Camera Translation

Estimate camera B's position in the coordinate frame of camera A.



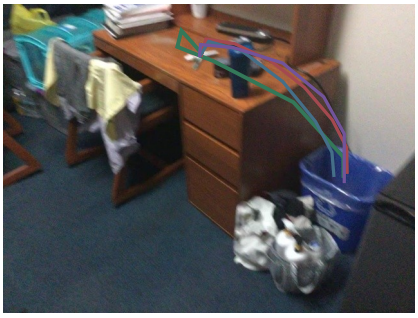
View 1 View 2

Model	Translation (cm)
GPT-6 Astra	[8, 0, -10]
GPT-6 Sol	[10, -10, 0]
Fable 5.1	[6, -3, 0]
Gemini 3.8 Flash	[25, 12, 0]
Qwen 3.8 Max	[10, 4, 0]
Qwen 3.5 397B	[15, 8, 5]
Qwen 3.5 27B	[8, 3, -12]
Opus 5.5	[15, -5, 0]

Reference translation: [12, 9, 0] cm

(c) TraceSpatial | 3D Trajectory

Predict a trajectory from the toothpaste to the blue waste bin.



GT Astra Fable Opus

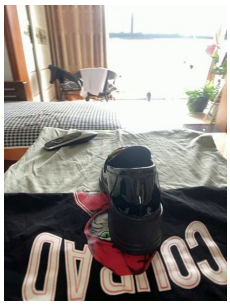
Image projection of 3D paths

Model	3D Fréchet (m)
GPT-6 Astra	0.525
GPT-6 Sol	0.634
Fable 5.1	0.826
Gemini 3.8 Flash	1.148
Qwen 3.8 Max	0.699
Qwen 3.5 397B	0.715
Qwen 3.5 27B	0.497
Opus 5.5	0.549

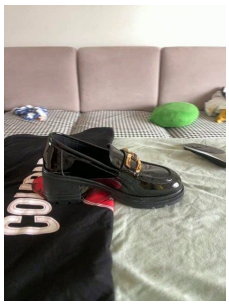
Figure A4. Geometry: displacement, reference frames, and trajectories. First-run predictions for three selected cases. (a) The jug moves right by 2 cm relative to the table; predictions distinguish movement, direction, and distance. (b) Camera translation is expressed in camera A's frame, with reference $[12, 9, 0]$ cm; orientation is omitted from this panel. (c) TraceSpatial paths are projected into the image, while the reported Fréchet distance is measured in 3D meters. The reference and Astra, Fable, and Opus paths are drawn; all eight individual errors are listed. These cases probe different geometric requirements, rather than interchangeable measures of task success.

(a) MindCube | Viewpoint Change

In which direction did the camera move between these views?



View 1



View 2

Model	First-run answer
GPT-6 Astra	D
GPT-6 Sol	B
Fable 5.1	A
Gemini 3.8 Flash	A
Qwen 3.8 Max	A
Qwen 3.5 397B	C
Qwen 3.5 27B	A
Opus 5.5	A

Reference A: diagonally forward and right

(b) MMSI-Bench | Relative Direction

If the open cupboard faces north, where is the cooking area?

View 1

View 2

Model	First-run answer
GPT-6 Astra	A
GPT-6 Sol	A
Fable 5.1	A
Gemini 3.8 Flash	D
Qwen 3.8 Max	C
Qwen 3.5 397B	B
Qwen 3.5 27B	B
Opus 5.5	D

Reference A: northwest

(c) 3DSRBench | Object Orientation

Which side of the person is facing the camera?

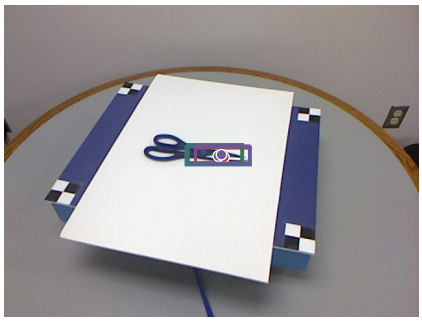
Model	First-run answer
GPT-6 Astra	B
GPT-6 Sol	B
Fable 5.1	C
Gemini 3.8 Flash	B
Qwen 3.8 Max	B
Qwen 3.5 397B	C
Qwen 3.5 27B	B
Opus 5.5	C

Reference B: left

Figure A5. Spatial Reasoning: errors depend on the reference frame. Selected first-run answers without YOLOE. (a) MindCube asks about camera movement between two views: A is diagonally forward-right, B right, C left, and D diagonally forward-left. (b) MMSI-Bench asks for a direction anchored to the open cupboard: A is northwest, B southwest, C north, and D northeast. (c) 3DSRBench asks which side of the person faces the camera: A is front, B left, C back, and D right. Green letters match the source reference; red letters do not. Different agents succeed on different frame transformations.

(a) UMD | Cutting

Locate the cutting region of the scissors.



GT box Astra Fable Opus

Reference box and first-run predictions

Model	Contact / box IoU
GPT-6 Astra	inside / 0.636
GPT-6 Sol	inside / 0.694
Fable 5.1	inside / 0.688
Gemini 3.8 Flash	outside / 0.000
Qwen 3.8 Max	inside / 0.706
Qwen 3.5 397B	inside / 0.540
Qwen 3.5 27B	outside / 0.000
Opus 5.5	inside / 0.734

(b) ReasonAff | Pulling

Locate the functional region to pull.

GT box Astra Fable Opus

Reference box and first-run predictions

Model	Contact / box IoU
GPT-6 Astra	inside / 0.197
GPT-6 Sol	inside / 0.150
Fable 5.1	inside / 0.103
Gemini 3.8 Flash	outside / 0.000
Qwen 3.8 Max	inside / 0.193
Qwen 3.5 397B	outside / 0.000
Qwen 3.5 27B	outside / 0.000
Opus 5.5	inside / 0.156

(c) RAGNet-3DOI | Two-Hand Interaction

Locate the region for the task “free using two hands”.

GT box Astra Fable Opus

Reference box and first-run predictions

Model	Contact / box IoU
GPT-6 Astra	outside / 0.000
GPT-6 Sol	outside / 0.000
Fable 5.1	inside / 0.906
Gemini 3.8 Flash	outside / 0.045
Qwen 3.8 Max	outside / 0.000
Qwen 3.5 397B	outside / 0.000
Qwen 3.5 27B	outside / 0.000
Opus 5.5	inside / 0.942

Figure A6. Affordance: usable contact and region coverage can disagree. First-run outputs on UMD, ReasonAff, and RAGNet-3DOI. Green rectangles are reference boxes; colored points and boxes show Astra, Fable, and Opus predictions. The table includes all eight models: contact is inside/outside the reference mask, and box IoU is intersection over union with the reference rectangle. Out-of-image points count as outside. The scissors and pulling examples contrast contact localization with region coverage; the two-hand example illustrates disagreement over the relevant functional region. Short task descriptions paraphrase the input instructions without changing the target.

(a) ALFWorld | Object and Light

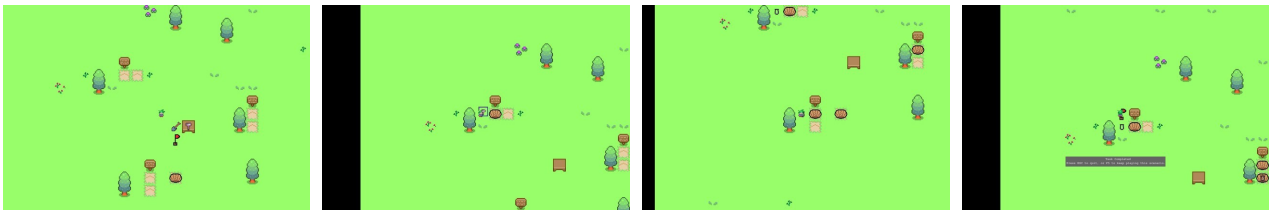
Turn on a lamp while holding the requested object.

obs. 1 obs. 3 obs. 6 obs. 8

Claude Opus 5.5 Reported success | 8 code rounds | 80.9 s

(b) DiscoveryWorld | Archaeology

Date the artifacts and flag the site containing the oldest one.

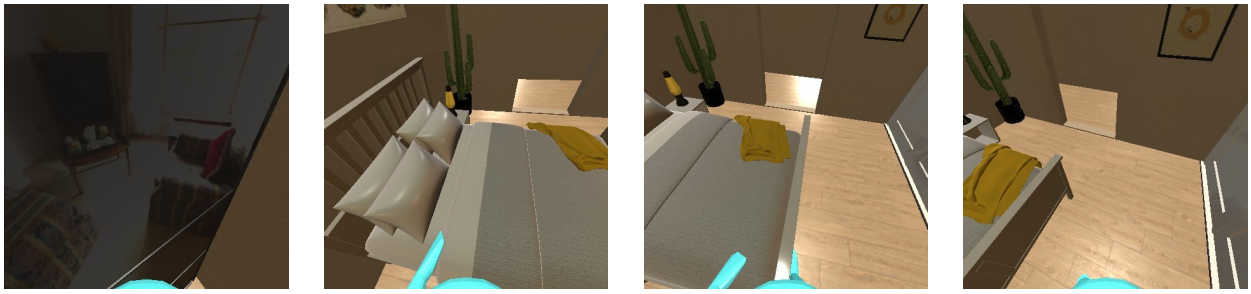


obs. 1 obs. 13 obs. 31 obs. 44

Claude Opus 5.5 Reported success | 7 code rounds | 167.9 s

(c) HumanCLAW | Sitting

Find the couch, approach until touching it, and sit.



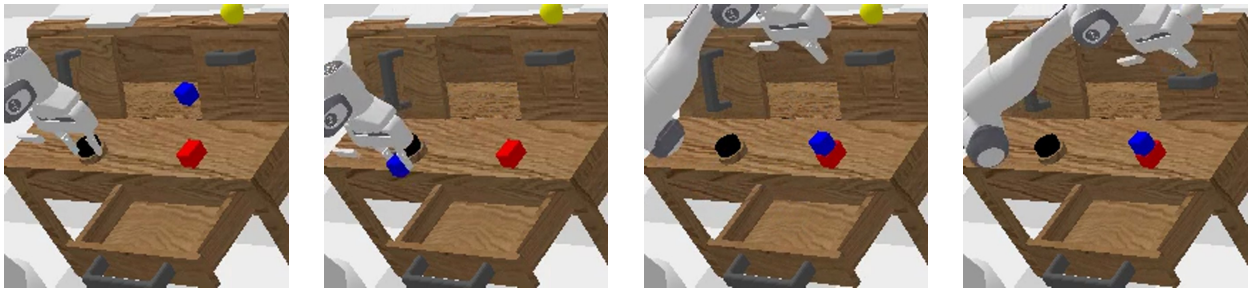
obs. 1 obs. 20 obs. 49 obs. 69

Claude Opus 5.5 Task failure | 70 code rounds | 933.6 s

Figure A7. Task Planning: a terminal episode need not be a completed task. Additional Opus 5.5 runs. (a) ALFWorld reports success after 8 code rounds under ceilings of 20 minutes, 8 rounds, 120 environment steps, 540 primitive calls, and 320k tokens. (b) DiscoveryWorld reports success after 7 rounds under 20 minutes, 8 rounds, 100 steps, 400 calls, and 320k tokens; its text/JSON interaction is illustrated with environment snapshots. (c) HumanCLAW stops after 70 rounds (933.6 s) without completing the couch-sitting task, under ceilings of 180 minutes, 101 rounds, 100 steps, 400 calls, and 2M tokens. The last case is a failed terminal state, not a wall-clock timeout. Frames are sampled saved observations, not continuous video.

(a) CALVIN | Sequence 0526

Complete the five instructed manipulation subgoals in sequence.

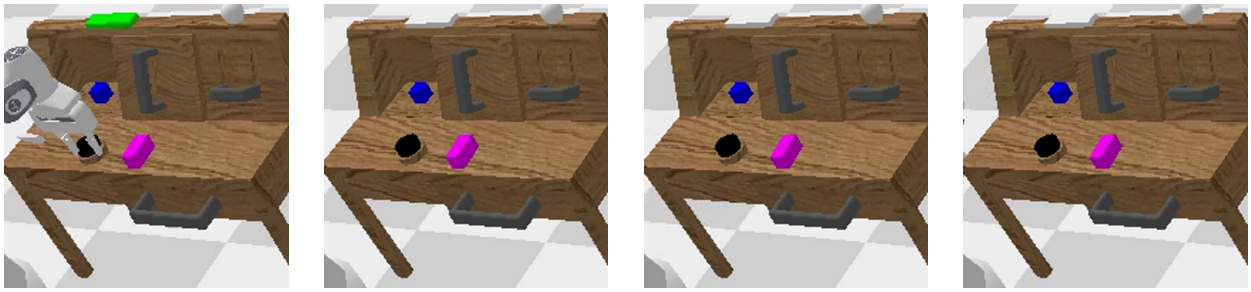


0.0 s 4.8 s 12.0 s 16.8 s

Claude Opus 5.5 Reported success | 40-round ceiling | 229.4 s

(b) CALVIN | Sequence 0557

Complete the five instructed manipulation subgoals in sequence.

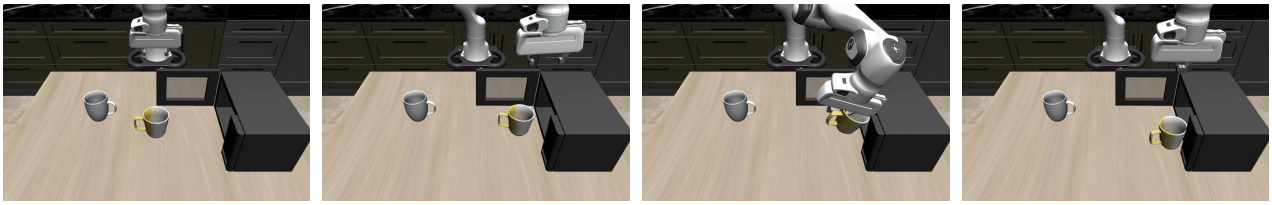


0.0 s 59.7 s 149.2 s 208.9 s

Claude Opus 5.5 Round limit reached | 40 rounds | 978.6 s

(c) LIBERO-PRO | Mug and Microwave

Put the yellow-and-white mug in the microwave and close it.



0.0 s 70.8 s 176.9 s 247.7 s

Claude Opus 5.5 Round limit reached | 24 rounds | 1308.6 s

Figure A8. Manipulation: motion can continue without completing all goals. Additional Opus 5.5 runs under RGB-expanded source interfaces. (a) CALVIN sequence 0526 reports success; (b) sequence 0557 exhausts 40 model rounds. Both allow 7,000 primitive calls and 1.28M tokens. (c) LIBERO-PRO requires placing a mug inside the microwave and closing it; the run exhausts 24 model rounds, with ceilings of 256 primitive calls and 640k tokens. Outcomes come from the terminal reports. Frame labels index the encoded recording, whereas the seconds beside each outcome are wall-clock runtime. Different task instances and source interfaces are illustrated here, not a matched comparison of models.